

EVALUATING ACCUFIX HEAD AND NECK DEVICE SHOULDER IMMOBILIZATION FOR
HEAD AND NECK RADIATION THERAPY

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Abstract

The field of radiation therapy has seen significant advancements in treatment precision, often involving various immobilization devices for patient positioning and motion monitoring. However, the effectiveness of shoulder immobilization systems, particularly in the neck and shoulder regions, require further investigation due to conflicting results and limited studies. This study aims to evaluate the accuracy and effectiveness of the Accufix™ Head and Neck Device shoulder cantilever depression system in reducing interfractional and intrafractional movement in head and neck cancer (HNC) patients. Patient data was collected for 3 larynx cancer patients and 1 tongue cancer patient undergoing volumetric-modulated arc therapy (VMAT) radiation therapy at Juravinski Cancer Centre. Patients were immobilized using a conventional head and neck thermoplastic mask and an Accufix Head and Neck device. AlignRT, an optical surface monitoring system (OSMS), was utilized to track real-time body surface movements during treatment in 3 translation directions (AP, SI, and LR) and 3 rotations (pitch, roll, and yaw). Interfractional shoulder positioning discrepancies were evaluated by conducting cone-beam computed tomography (CBCT) scan and planning computed tomography (CT) scan image registration in two anatomical locations: the target volume (T-IM) and the mid-clavicles (C-IM). Intrafractional motion across all patients remained low for both translational and rotational shifts, with 6.42% exceeding a 5mm margin and 0.02% exceeding a 3° margin, respectively. Differences between target TV-IM and C-IM remained within +/- 3mm of shifts for the majority of fractions. Little consistency was found between AlignRT data and C-IM data, with shifts ranging from 10mm to -5mm, attributed to the surface geometry and shape of the region of interest (ROI) we tracked. While shoulder immobilization using the Accufix system was found to be sufficient, AlignRT's accuracy in reproducing patient shoulder positioning was limited in our study. Factors influencing surface-guided systems, such as ROI size and location, need careful evaluation. Further studies are needed to validate these findings and explore potential improvements.

Introduction

Background

Cancer treatment has progressed exponentially in the past century, with the availability of chemotherapy, radiation therapy, immunotherapy, and surgery coming to light. Around 60% of patients will receive radiotherapy during the course of their treatment (Chierchini et al., 2019), yet this accounts for only about 5% of the total cost of cancer care (Baskar et al., 2012). Alongside other modalities, this treatment has proved to be effective and essential for both treating cancer and palliative care. Irradiation is achieved by the deposition of photons into the body using linear accelerators, thus resulting in the ionization of biomolecules. This targeted dose of radiation damages the DNA within cells, directly and indirectly, consequently eliciting various cellular responses, many of which often lead to cell death (Zeman et al., 2020). Treatment planning, patient positioning, and monitoring are critical elements that each play a role in ensuring a high degree of precision in targeting cancerous cells while minimizing radiation to healthy, surrounding regions. Patient positioning can be quantified between fractions or doses (interfractional), as well as during treatment itself (intrafractional). Immobilization devices play an important role in both interfractional and intrafractional movement and can vary from thermoplastic masks, shoulder depression systems, vacuum bags, or body rests. Treatment approaches commonly

involve employing diverse combinations of these devices. However, the optimal technique for maximizing dose accuracy while minimizing movement is contingent upon the section of the body evaluated (Malicki, 2016).

Shoulder Devices

The use of various thermoplastic masks, head supports, and neck supports seem to have limited effects on movement in the lower neck and supraclavicular regions (Gilbeau et al., 2001; Lin et al., 2017; Zhang et al., 2006), save for an E-frame mask implemented by Fukao et al. (2020). To combat shoulder movement in head and neck cancer (HNC) patients, shoulder devices such as cantilever depression systems, vacuum cushions, and rubber pull systems have been introduced. In a comparison between three systems: a head and shoulder five-point thermoplastic mask; a head-only mask with a vacuum neck support and shoulder locks fixed to a cantilever board; and a head and shoulder eight-point thermoplastic mask with a vacuum cushion; the five-point mask system exhibited the least random errors in all rotational directions and fewer translational errors compared to the shoulder lock system (Hansen et al., 2014). In a study for helical tomotherapy, HNC patients immobilized using a head and shoulder thermoplastic mask and rubber traction pulls were evaluated for setup margins. Setup margins of 5mm were found to be sufficient for errors in translational and rotational directions for HNC patients, suggesting these immobilization methods are sufficient (Thondykandy et al., 2015). However, without comparison studies, this cannot be inferred.

Regional Variations

Measurements of interfractional movement are typically achieved through image registration of the planning kVCT and the CBCT using bony anatomy surrounding the target volume as a reference (Ove et al., 2012). Previous studies have found that lower neck structures experienced higher setup discrepancies when aligning to C2 as opposed to the lower targets, with the largest variation in the AP and SI directions (Neubauer et al., 2012; Ove et al., 2012). It has also been suggested that patients with lower isocenters experienced greater changes in cervical spinal angles over time (Garg et al., 2004). Studies comparing headrests in conjunction with thermoplastic masks indicate less effective immobilization in the neck region, regardless of the headrest type used, attributed to the semi-independent movement of the skull with respect to the neck (Ahn et al., 2009; Lin et al., 2017). Thus, greater setup margins for target volumes in proximity to the shoulder level have been suggested to combat these deviations (Rotondo et al., 2008) due to significant dose perturbations as treatment progresses (Ballivy et al., 2006). Neubauer et al. (2012) indicate lower neck coverage is greatly affected by superior shifts and larger posterior shifts, where up to 1 Gy can be lost from shoulder variation over the course of an individual's treatment. Adjusting to larger planning target volume (PTV) margins is advised for neck region targets to increase dosimetric coverage and account for suboptimal immobilization of lower neck regions.

Intrafractional motion is typically monitored for stereotactic body radiation therapy (SBRT) using surface or image-guided techniques (Freisleder et al., 2020). A small number of studies have investigated the impact of ROI location on intrafractional motion relative to the isocenter, however results have been inconclusive. Translational errors have been found to vary in both upper and lower ROIs as a result of rotational shifts, with pitch being reported with the largest systematic and random errors (Nakata

et al., 2013; Zhang et al., 2006). An investigation (Kang et al., 2020) compared head-only mask usage with and without a vacuum bag, noting increasing roll and pitch displacements after 10 minutes and yaw, SI direction, and LR direction displacements after 20 minutes, similar to a past suggestion of a time trend (Zhang et al., 2006). LR shifts exceeded a PTV margin of 2mm, however, the authors were unable to conclude whether the upper or lower spine regions are better immobilized. It is evident that shifts in the neck region are larger than expected during treatment.

Current Objectives

The current literature suggests that thermoplastic masks are equally sufficient in immobilization for the head region, however, shifts commonly occur in the neck and shoulder region below the mandible. Shoulder systems have been implemented to reduce motion beyond these masks, although there is little evidence to suggest that they are effective. Further research has yet to be conducted in analyzing these shoulder systems relative to one another, particularly in HNC patients. In addition, intrafractional motion studies regarding the lower neck and shoulder areas are limited in quantity, with conflicting results. Dosimetric coverage of target volumes is influenced by these shifts as well, yet in many instances, deviations from planning images exceed that of PTV and setup margins. Institutional margins should be evaluated with respect to the accuracy of their immobilization devices. To address some of these gaps, we sought to evaluate the accuracy and effectiveness of the shoulder cantilever depression system for reduction in interfractional and intrafractional movement in larynx and tongue cancer patients. Through the use of AlignRT, a clavicle-based ROI will be tracked to measure 6D deviations in positioning. The accuracy of AlignRT for intrafractional motion and tattooless setup has been validated by numerous studies. Despite this, little research has been conducted on the use of AlignRT for larynx cancer patients. Thus, the accuracy of AlignRT's Optical Surface Monitoring System (OSMS) about the lower neck and supraclavicular region will also be investigated.

Methodology

Treatment and Monitoring Setup

Patient data was acquired for OSMS setup for 3 larynx cancer patients (Patients A, B, & D) and 1 tongue cancer patient (Patient C) undergoing VMAT with the Truebeam STx linear accelerator system (Varian, Palo Alto, California) at Juravinski Cancer Centre with either 20 or 30 fractions (200-255 cGy/fraction) beginning in October 2023. The inclusion criteria were: 1) diagnosis of larynx or tongue malignancies; 2) undergoing VMAT radiotherapy. Patients undergoing palliative treatment or patients with difficulties collecting OSMS data were excluded from this study. Patients were informed and asked for consent in their participation in the study, then asked to lower or remove their shirt or gown to expose the supraclavicular region. Patients were positioned using conventional tattoo and laser-based setup and immobilized using a thermoplastic mask covering the head and neck (Aquaplast Corp., Wyckoff, NJ) and the AccuFix™ Head and Neck Device (Qfix, Avondale, PA), as in Figure 1.

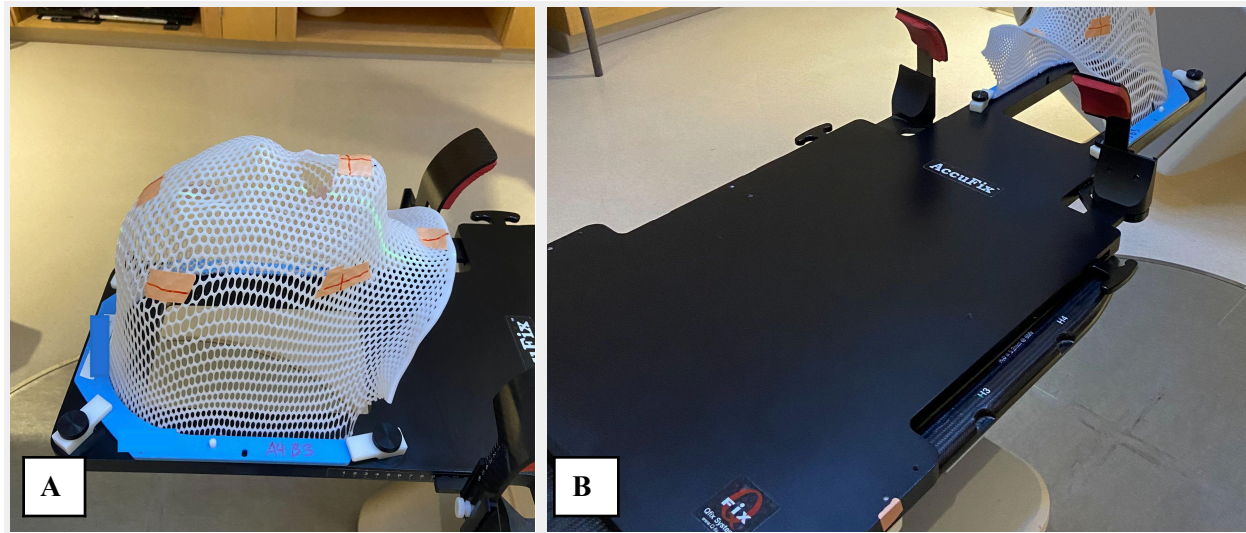


Figure 1. Photograph of the immobilization devices used throughout the study. A) Aquaplast thermoplastic mask; B) Accufix cantilever shoulder depression system.

Optical Surface Monitoring

Data was exported from the ARIA Oncology Information System (Varian, Palo Alto, California) to computers integrated with the AlignRT's OSMS system (VisionRT, London, UK), consisting of 3 cameras and projectors which measure real-time body surface movement information in 3 cardinal translations and 3 axis rotations. AlignRT matches the current position to the reference position, which can be a surface imported from DICOM or captured using AlignRT, and halts the radiation beam when it detects that patient motion has exceeded a certain threshold. This is achieved by projecting a speckle pattern upon the user-defined region of interest (ROI), calculating the translations and rotations necessary to match the surfaces through a rigid body registration algorithm (Bert et al., 2005). The output consists of log files containing Real-Time Deltas (RTDs) displaying the delta shifts in a vertical or anterior-posterior (AP) shift, a longitudinal or superior-inferior (SI) shift, and a lateral or left-right (LR) shift, as well as rotations about the treatment isocenter for pitch, roll, and yaw. These parameters (as well as various other parameters such as total translation, amplitude, beam state, and root mean squared) are recorded each time a shift is detected, thus having varying time intervals of about 7 seconds. At our institution, the threshold of patient motion for clinical treatment is set to 3.0mm and 3.0° for intrafractional motion monitoring. However, for the purpose of measuring interfractional motion, reference positions were not captured at the beginning of treatment, rather movement was monitored with respect to the DICOM planning CT. This requires a larger range of motion to be measured, thus threshold ranges were increased to a maximum of 9.9mm and 9.9° for translational deviations and rotational deviations, respectively, while the beam hold timer was increased to 5 seconds. Automatic beam hold gating was not utilized. Previous studies have suggested that the region of interest (ROI) should remain stable, avoiding the axillary and lymph drainage regions, and maintaining a sufficient gap between the patient's mask edge and the ROI (Bellala et al., 2023; Sauer et al., 2021). Additionally, smaller ROIs are found to be more accurate for tumor tracking, whereas larger ROIs are preferable for monitoring positional deviations (Al-Hallaq et al., 2022; Bellala et al., 2023; Psarras et al., 2023, Sauer et al., 2021). Thus, an area just below the

supraclavicular region illustrated in Figure 2, highlighted in white, was tracked using AlignRT during patient setup and treatment.

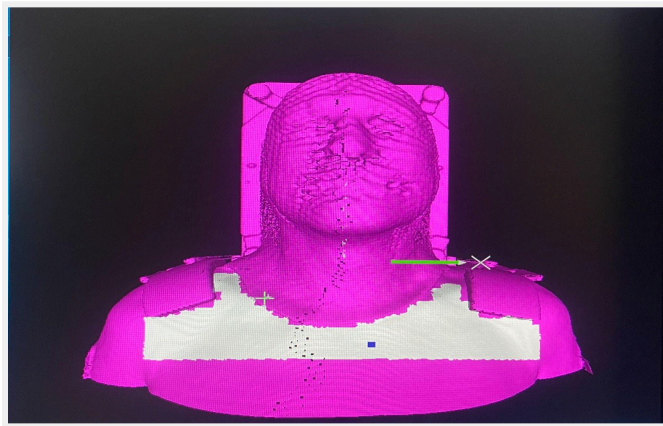


Figure 2. Screen capture of a 3D render of a patient's body in purple on Varian software. The ROI for AlignRT tracking was mapped to be the white portion.

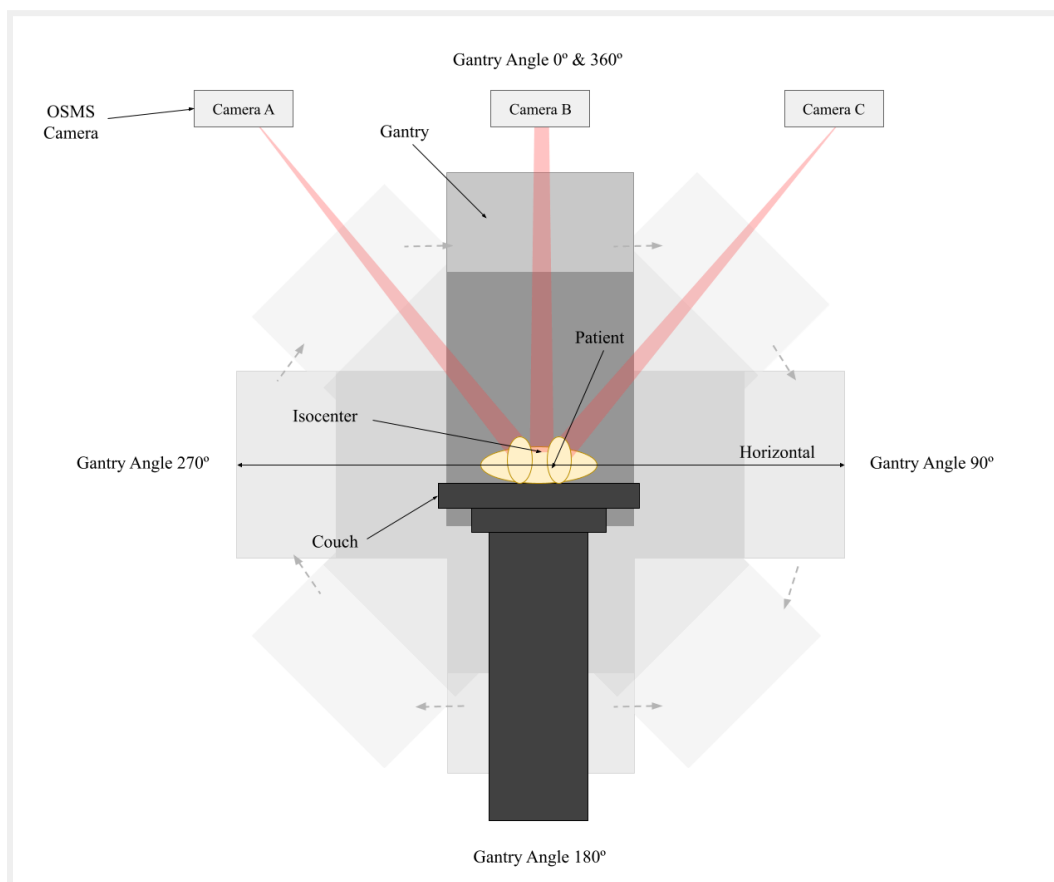


Figure 3. Schematic diagram of gantry angles and OSMS camera positioning during VMAT. The optical surface monitoring system incorporates three cameras designated as Cameras A, B, and C. Each camera is mounted to the ceiling. Camera B is offset from the plane of the image towards the viewer. The gantry cannot rotate beyond 180° from the 0° or 360° position.

AlignRT was used for the duration of the treatment, then the reported log files were exported in the form of .txt files. This raw data was then exported to a .csv file for statistical analyses and graphing in R. At our institution, gantry angles above the horizontal of 0° to 90° and 270° to 360° blocked optical surface cameras A, B, and C, as depicted in Figure 3, determined through patient treatment data in ARIA. Data falling under these angles were skewed or appeared as “N/A” (Appendix 1), and thus were removed. Intrafractional data was gathered by averaging the initial 10 points of OSMS data while the treatment beam was active, then subtracting the following points by that value to determine the positional variance from the start position during treatment. This allows us to ensure each patient’s breathing cycle is filtered out. For interfractional data, the mean of the initial 10 points of OSMS data where the treatment beam was inactive was calculated.

CT Image Registration

Planning CT scans were acquired using the SOMATOM Definition AS system (Siemens Healthineers, Erlangen, Germany). Prior to each patient’s treatment fraction, kVCBCT were performed using the Truebeam STx. During treatment, kVCBCT and planning CT scans were registered by radiation therapists to match mid-neck anatomy, surrounding the target volume. If a re-CBCT was acquired that treatment fraction, the second CBCT was used. Calculation of positioning using this method will be denoted as target volume-based image matching (TV-IM). To assess variability in shoulder position, automatic kV image rigid body registration was performed on the same image sets using Eclipse, focusing on mid-clavicle alignment under an intensity range of 200 to 1700 Hounsfield units (HU) to focus on bony anatomy, as pictured in Figure 4. Manual corrections were employed if automated alignment was visually unsatisfactory. Calculation of positioning using this method will be denoted as clavicle-based image matching (C-IM).

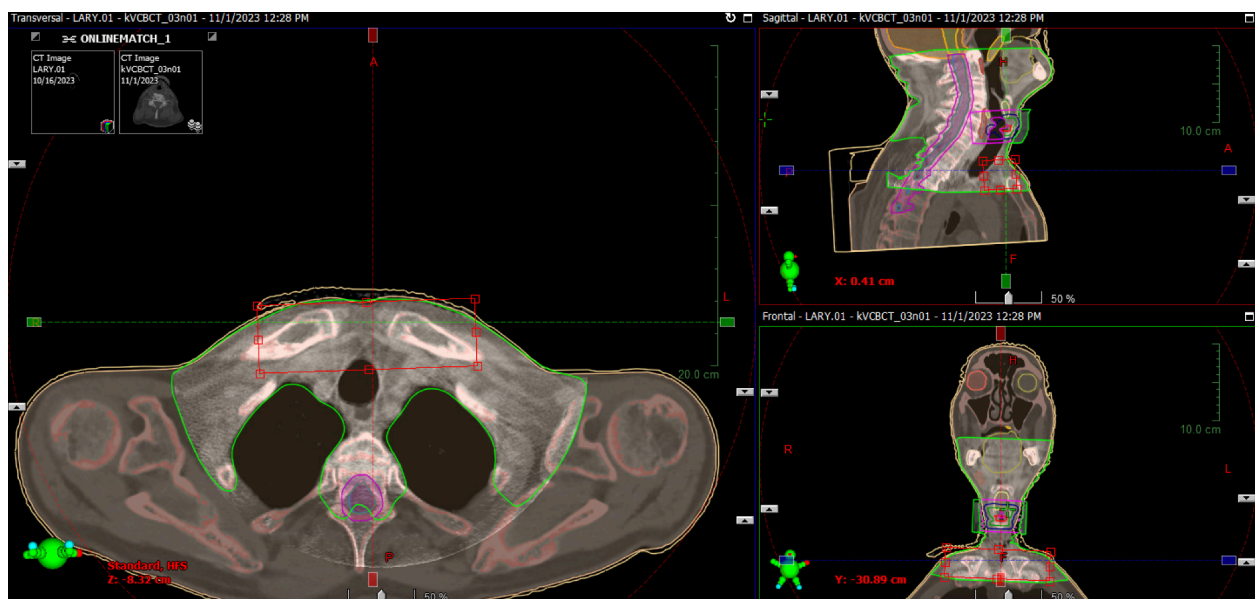


Figure 4. Screen capture of clavicle-based image registration (C-IM) executed in Eclipse software. Bony anatomy is aligned within the volume of interest (VOI) designated by the red box.

Results

AlignRT datasets monitoring shifts where the radiation beam was active for each patient and fraction utilized for further analyses are displayed in Appendix 2. Histogram plots (Appendix 3) indicate the data is not consistently normally distributed across either translational or rotational shifts in all patients. Data was grouped into clusters based on connecting gantry angle positions observed during both arcs of a patient's treatment. Cluster 1 is characterized by the gantry moving clockwise from 180° to 270°. Cluster 2 is defined by the gantry's clockwise movement from 90° to 180° and counter-clockwise movement from 180° to 90°. Cluster 2 defines gantry movement counter-clockwise from 270° to 180°. The angles within each cluster correspond to the same time frame, with separation between the two clusters occurring during intervals when the gantry is positioned above the horizontal, obstructing the view of the cameras (Figure 3). Cluster analyses (Appendix 4) show that organizing the data by date is sufficient, as each cluster remains fairly similar with no striking differences across each type of shift and patient.

Intrafraction Motion

Intrafraction motion measurements are displayed in Tables 1 and 2 between the position at the beginning of each treatment fraction and AlignRT recorded data points. Table 1 exemplifies that motion in the AP, SI, and LR directions typically remain within a 5mm margin. The mean percentage of data points in all fractions across all patients that were displaced beyond 5mm during treatment in the AP, SI, and LR directions were 0.01%, 1.92%, and 1.77%, respectively. Some shifts occurred in the SI and LR directions exceeding 3mm, with the mean percentage of data points displaying beyond 3mm in the AP, SI, and LR directions were 0.99%, 11.05%, and 7.23%, respectively. Table 2 exemplifies that pitch, roll, and yaw rotations almost always remained within a margin of 3°. The greatest intrafraction rotations are observed in the pitch and yaw rotations, with 8.86% and 6.63% of data points across all patients and treatment fractions exceeding 1°. 2.34% of roll rotations exceeded 1° across all data points.

Table 1. Percentages of translational shifts out of all treatment fractions exceeding 3mm and 5mm for each patient. Shifts are measured relative to the position at the beginning of treatment, following kVCBCT and delta couch shifts. Shifts analyzed included the anterior–posterior (AP), superior–inferior (SI), left–right (LR) directions. Each data point refers to a detection of movement by AlignRT data, recorded about every 7 seconds during an active treatment fraction. Data is sampled exclusively from when the gantry is at or below the horizontal (as depicted in Figure 3).

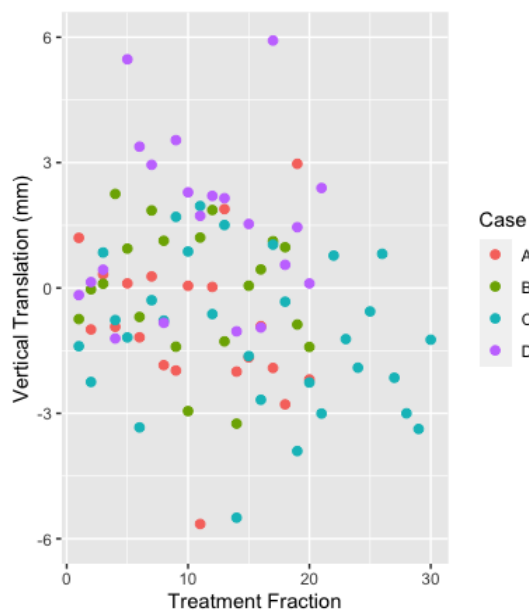
	AP (mm)		SI (mm)		LR (mm)	
	> 3mm	> 5mm	> 3mm	> 5mm	> 3mm	> 5mm
A	0.706	0	32.745	7.502	5.649	2.736
B	1.656	0.040	0.767	0	13.772	3.635
C	1.611	0	10.274	0.196	4.990	0.294
D	0	0	0.403	0	4.497	0.403

Table 2. Percentages of rotational shifts out of all treatment fractions exceeding 1° and 3° for each patient. Shifts are measured relative to the position at the beginning of treatment, following kVCBCT and delta couch shifts. Shifts analyzed included pitch, roll, and yaw rotations measured about the treatment isocenter. Each data point refers to a detection of movement by AlignRT data, recorded about every 7 seconds during an active treatment fraction. Data is sampled exclusively from when the gantry is at or below the horizontal (as depicted in Figure 3).

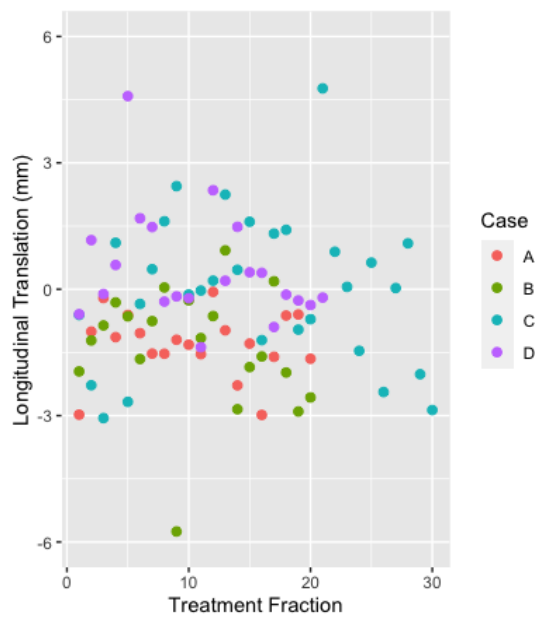
	Pitch (°)		Roll (°)		Yaw (°)	
	>1°	>3°	>1°	>3°	>1°	>3°
A	2.118	0	0.530	0	4.678	0.618
B	16.922	0.121	5.614	0.081	13.651	1.575
C	7.926	0	0.196	0	5.186	0
D	8.456	0	3.020	0	3.020	0

Interfractional Motion

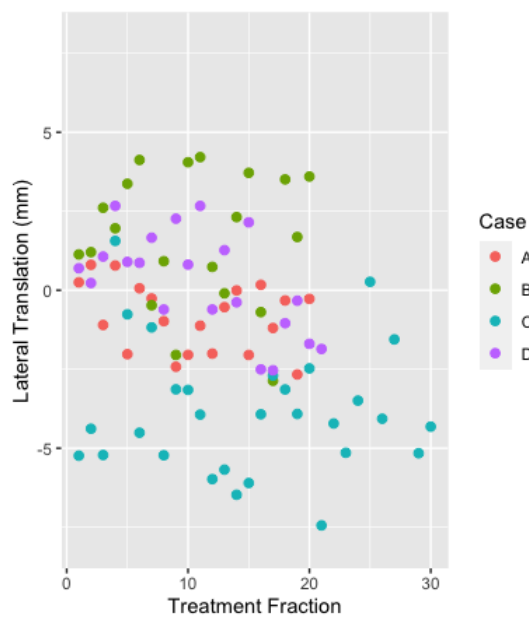
Differences between TV-IM and C-IM in the AP, SI, and LR directions, as well as the pitch, roll, and yaw rotations across the course of the patients' treatment were analyzed in Figure 5 and Figure 6. These values refer to the offsets in clavicle positioning when the target volume is aligned to the planning CT position. The middle 50% of fractions across all patients remain within +/- 3mm of AP, LR, and SI discrepancies, however a larger degree of variation is observed in Patient C, or Case C. Clavicle offsets in the LR direction were larger than AP and SI offsets across all patients, with Patient C exceeding a difference of 5mm in 37.04% of fractions. Pitch, roll, and yaw discrepancies of the middle 50% of fractions remained within +/- 3°, however Patient C had slightly greater rotational offsets between TV-IM and C-IM.



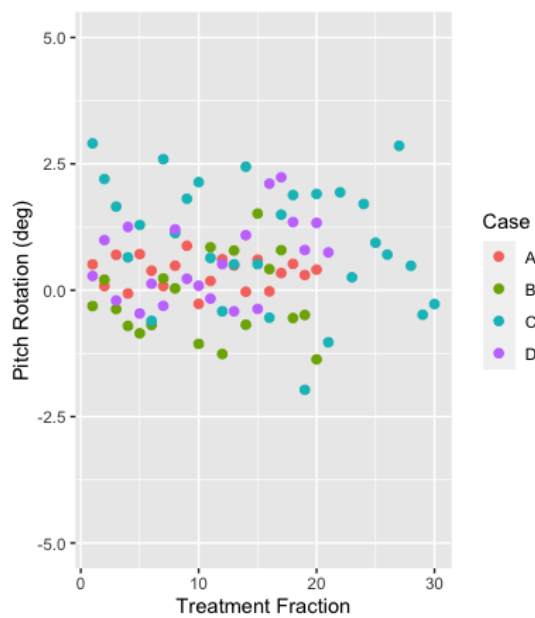
a)



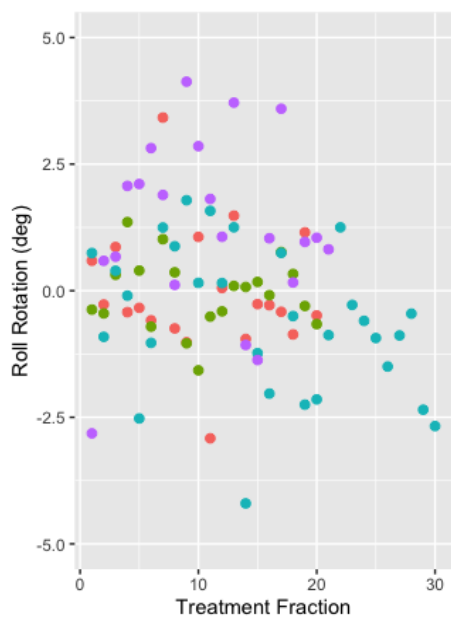
b)



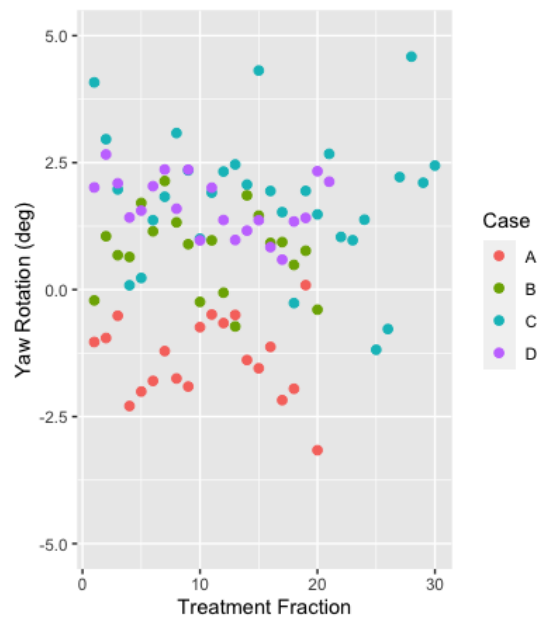
c)



d)



d)



f)

Figure 5. Difference in patient clavicle positioning across treatment fractions between TV-IM and C-IM datasets. Each patient is exemplified by a different case and colour. Each point represents the discrepancy between the intended clavicle position and the observed clavicle position for a specific treatment fraction. a) Difference in vertical (AP) translations across fractions; b) Difference in longitudinal (SI) translations across fractions; c) Difference in lateral (LR) translations across fractions; d) Difference in pitch across fractions; e) Difference in roll across fractions; f) Difference in yaw across fractions.

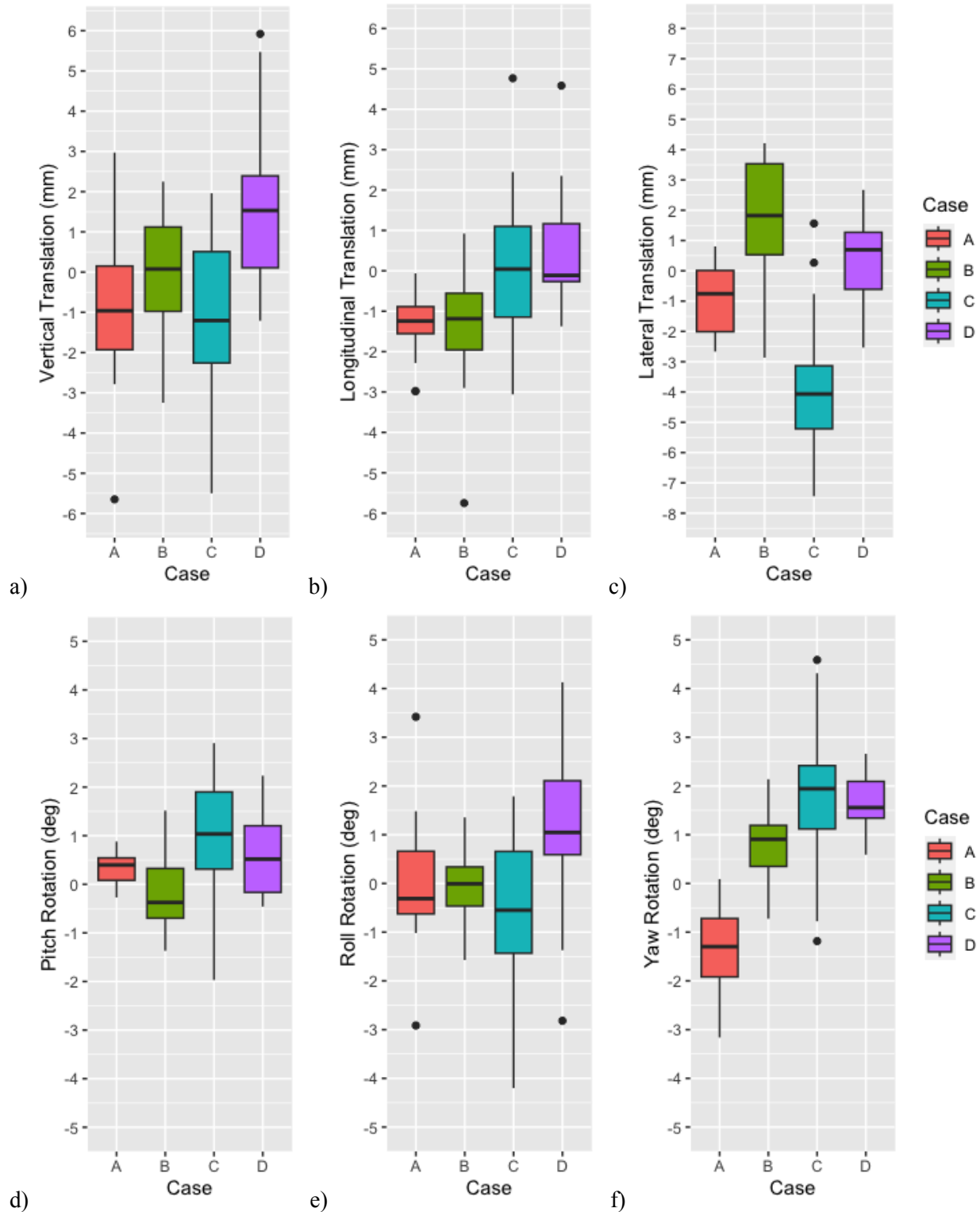
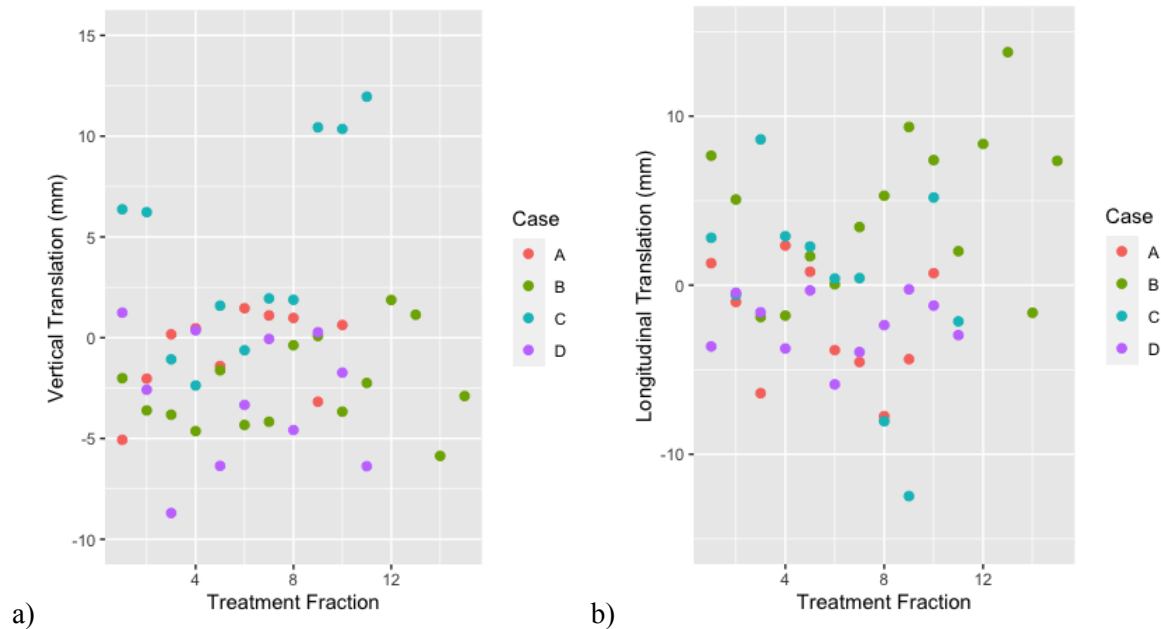


Figure 6. Translational and rotational differences in patient clavicle positioning between TV-IM and C-IM data. Each patient is exemplified by a different case and colour. Each point represents the discrepancy between the intended clavicle position and the observed clavicle position for a specific treatment fraction. a) Difference in vertical (AP) translations across fractions; b) Difference in

longitudinal (SI) translations across fractions; c) Difference in lateral (LR) translations across fractions; d) Difference in pitch across fractions; e) Difference in roll across fractions; f) Difference in yaw across fractions. Boxes represent the middle 50% of observations, with a line representing the median value. Outliers are represented by black circles.

AlignRT & Image Registration Consistency

Differences between C-IM and AlignRT data in the AP, SI, and LR directions, as well as the pitch, roll, and yaw rotations across the course of the patients' treatment were analyzed for Figure 7 and 8. These values refer to the offsets between clavicle positioning as measured by CT image registration and AlignRT surface monitoring data. The middle 50% of fractions across all patients ranged between +8mm and -5mm of AP, SI, and LR. The range of variation in the LR and SI directions was slightly larger than that of the AP direction. Results in the SI direction display the least accuracy of AlignRT to C-IM. Pitch, roll, and yaw discrepancies of the middle 50% of patients were found to range from $\pm 4^\circ$. The range of rotational discrepancies was largest in roll and yaw across all patients and fractions.



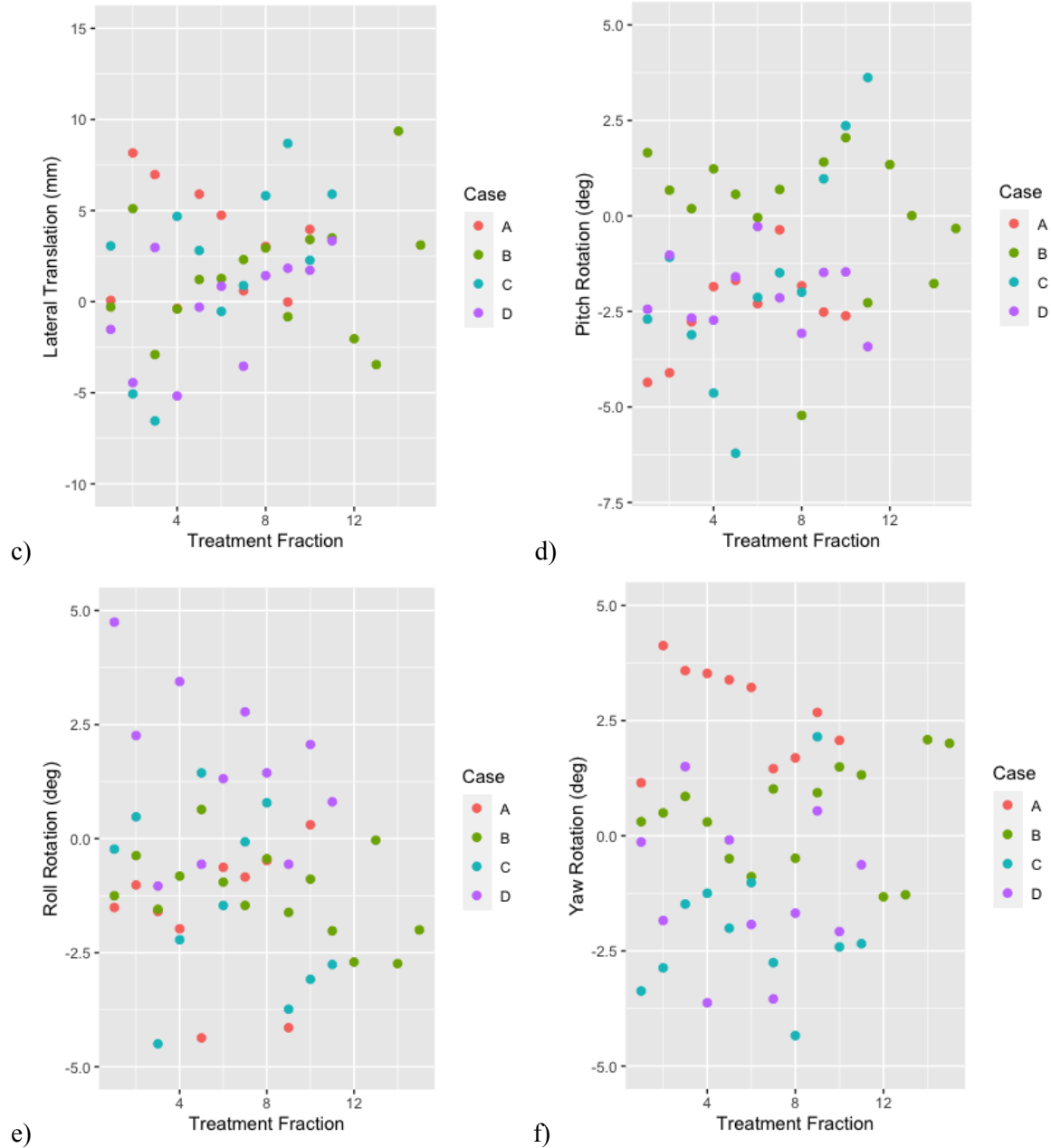


Figure 7. Difference between patient shoulder positioning recorded by image registration and recorded by surface monitoring across treatment fractions between C-IM and AlignRT datasets. Each patient is exemplified by a different case and colour. Each point represents the discrepancy between the observed clavicle position and the AlignRT-recorded clavicle position. a) Difference in vertical translations across fractions; b) Difference in longitudinal translations across fractions; c) Difference in lateral translations across fractions; d) Difference in pitch across fractions; e) Difference in roll across fractions; f) Difference in yaw across fractions. Correlation coefficients are displayed in Appendix 5.

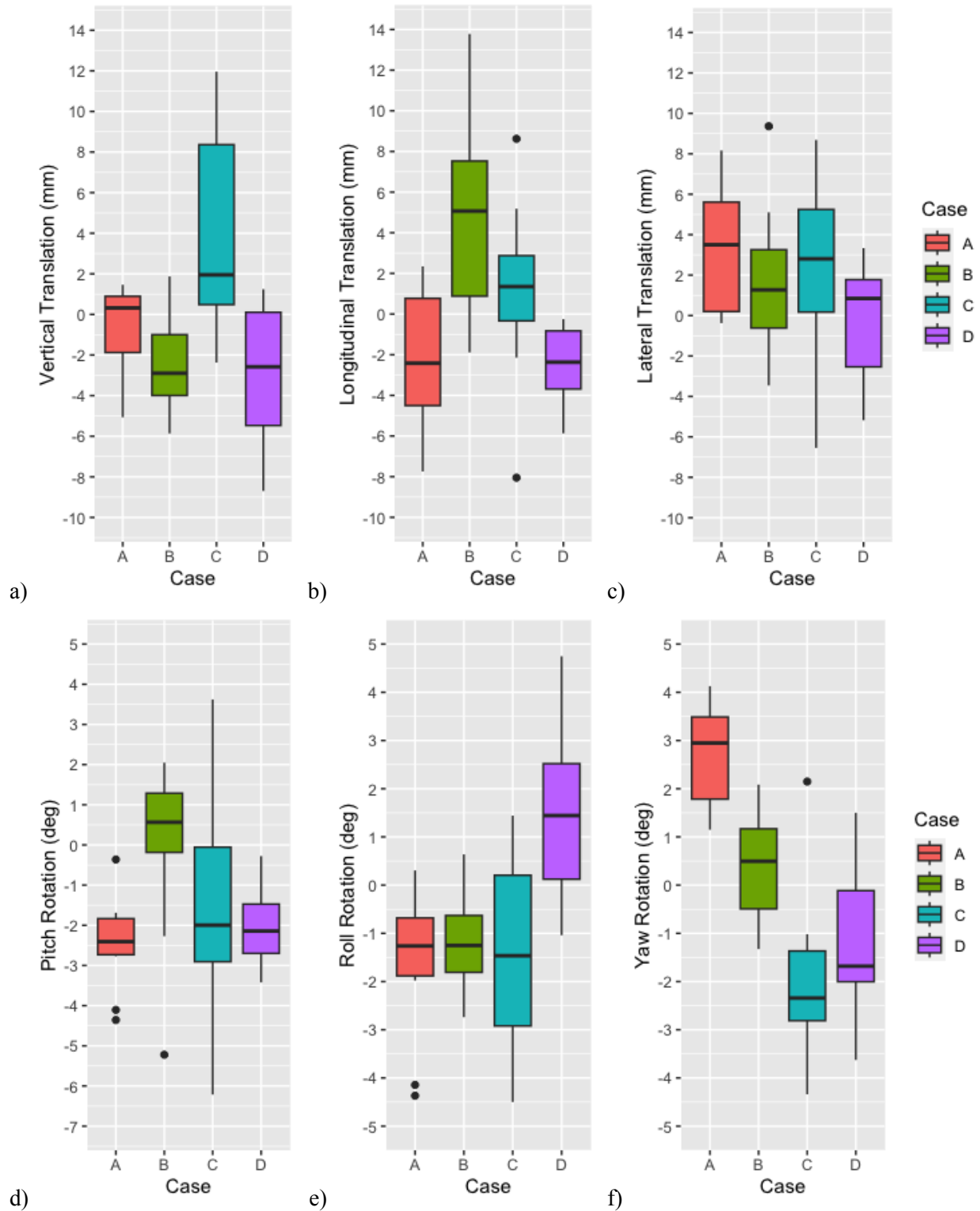


Figure 8. Translational and rotational differences in recorded patient clavicle positioning between C-IM and AlignRT datasets. Each patient is exemplified by a different case and colour. Each point represents the discrepancy between the clavicle position as determined by surface monitoring and clavicle position as determined by image registration for a specific treatment fraction. a) Difference in vertical

(AP) translations across fractions; b) Difference in longitudinal (SI) translations across fractions; c) Difference in lateral (LR) translations across fractions; d) Difference in pitch across fractions; e) Difference in roll across fractions; f) Difference in yaw across fractions. Boxes represent the middle 50% of observations, with a line representing the median value. Outliers are represented by black circles.

Discussion

Our investigation sought to evaluate movement of the shoulders via clavicle-based image registration and surface-guided tracking of the supraclavicular region when immobilized using the Accufix Head and Neck device. There is little literature to suggest shoulder cantilever depression systems are ineffective, but many default to head and shoulder thermoplastic masks as their standard of care (Hansen et al., 2014; Thondykandy et al., 2015). Intrafractional motion for translational shifts remained under 5mm in 93.58% of data points analyzed across all fractions while rotational shifts remained under 3° in 99.98% of data points across all fractions evaluated. The majority of interfractional movement remained within +/- 3mm, with the greatest being shifts in the LR direction. However, when evaluating the accuracy of AlignRT to clavicle-based image registration, the majority of shifts exceeded that of our expectations.

During treatment, patient movement is regulated by clinical immobilization devices, specifically thermoplastic masks and shoulder depression systems at our institution. Uncertainties such as technician errors, motion of the target, patient setup errors, and patient movements are often accounted for by the planning treatment volume (PTV): a margin of typically 5mm surrounding the target volume. Our results for intrafractional motion aligns with previous findings, wherein pitch, roll, and yaw had smaller magnitudes of shifts compared to translations, yet remained under 5mm across all 41 patients undergoing stereotactic ablative radiation therapy (Kang et al., 2020). However, many current intrafractional studies of the neck and shoulder regions utilize methods other than surface-guided techniques, such as live x-ray registration (Kang et al., 2020). Thus, there is little literature to discuss this aspect. Nevertheless, patient immobilization of the shoulders seems to be sufficient during treatment fractions.

Cone Beam Computed Tomography (CBCT) is often considered the gold standard for tracking interfractional movement in patients to guide repositioning prior to treatment (Carl et al., 2018). Prior to each treatment fraction, a pretreatment kVCBCT is conducted on the patient. This scan is then aligned with the planning CT scan, which provides guidance on the accurate positioning of the patient's interior anatomy for the treatment. Typically, matching kVCBCT scans to planning CT scans is achieved by automatically registering the bony structures surrounding the target volume. To determine the magnitude by which the shoulders move irrespective of the target volume, we analyzed interfractional motion organized both by fraction and by patient in a time series. Movement of the shoulders, specifically the clavicle area, often remained within a margin of +/- 3mm, with the largest variation in LR shifts. It has been suggested that the typical head and shoulder thermoplastic shell tightly conforms to the patient's anterior and lateral sides of the head, while the lack of constraint in the SI direction results in greater shifts (Lin et al., 2017). Our study evaluated the shoulder cantilever depression system, which lacks constraint on the lateral sides of the body. Thus, this could have resulted in greater shifts observed in the LR direction. In addition, the lack of a significant number of fractions with shifts in any direction exceeding 5mm or 5° aligns with previous findings with similar immobilization devices. In a study for helical tomotherapy (Thondykandy et al., 2015), HNC patients immobilized using a head and shoulder

thermoplastic mask and rubber traction pulls were evaluated for setup margins. Shoulder pulls function similarly to the Accufix Head and Neck device, in that the majority of its immobilization arises from restraining SI direction movement. Setup margins of 5mm were found to be sufficient for errors in translational and rotational directions for HNC patients, suggesting these immobilization methods are sufficient (Thondykandy et al., 2015). Alternatively, Rotondo et al. (2008) evaluated the use of a Type-S head and shoulder thermoplastic mask compared to a head-only mask with a shoulder depression system fixed to a cantilever board. No significant differences were found between the immobilization systems with respect to any direction or individual landmark, suggesting that both are sufficient in patient setup reproducibility. There was, however, a significant degree of variation in both translational and rotational shifts in Case C, the only patient analyzed with tongue cancer rather than larynx cancer. Both are immobilized in the same manner, and there is little positional variation between oropharyngeal and laryngeal cancer to suggest that this would make a significant difference upon shifts observed, however the distance from the clavicles to the isocenter of the tongue is slightly greater. Additionally, each of the larynx patients were treated by the same radiation therapists, however the tongue patient was treated by different therapists. Both reasonings could have played a role in larger positional discrepancies.

Interfractional discrepancies between observed clavicle positioning as determined by C-IM versus AlignRT data are large, often exceeding PTV margins. This challenges previous findings that AlignRT has submillimeter accuracy in patient setup and reproducing patient positioning (Carl et al., 2018; Mueller et al., 2023; Rudat et al., 2023; Song et al., 2022). An international sample of institutions displayed that about 50% use SGRT clinically (Batista et al, 2022), yet there remains some difficulties in the use of this radiotherapy technique, such as free breathing and ROI choice. Free breathing (Appendix 2) throughout treatment should not have played a significant impact in surface monitoring discrepancies as we calculated the mean of the first 10 points prior to treatment. Furthermore, respiration-induced uncertainties have been found to have little effect on typically larger setup uncertainties (Batin et al., 2016). The literature suggests that ROI size and location plays a significant role in AlignRT accuracy, however little evidence suggests the shape and size of our ROI utilized is ineffective. The size of our ROI was slightly larger than that of a typical HNC patient, yet smaller than those for pelvis or breast radiation therapy (Bellala et al., 2023). Drawbacks to a larger ROI than usual include a slower monitoring rate and decreased sensitivity to local anatomical changes, however both of these do not have an effect on tracking patient setup positioning (Al-Hallaq et al., 2022). A study on various ROI choices for breast cancer patient setup accuracies discussed that topographically salient features are necessary for accurate measurements of rotational shifts (Sauer et al., 2021). Specifically, ROIs with symmetrical or spherical structures (such as the breast's surface) have difficulty in the detection of translational shifts, however are particularly limited in the detection of rotational shifts. The symmetry of the clavicles could explain a difference in the resulting pitch, roll, and yaw shifts between C-IM and AlignRT. Dose distribution can vary significantly from even a small rotation of one degree, specifically in target volumes located far from the isocenter (Peng et al., 2011). Too flat or planar of a surface has been found to result in lateral and longitudinal inaccuracies (Al-Hallaq et al., 2022), which may have also played a role in significant SI and LR discrepancies between our C-IM and AlignRT datasets. An investigation of the effectiveness of optical surface imaging (OSI) compared to CBCT in HNC and breast cancer patients found that discrepancies may have arisen due to the difference between acquiring volumetric images compared to surface images (Li et al., 2021; Padilla et al., 2014). Particularly, flat images contain less information than volumetric images and could have resulted in lower than expected shifts, of which we found in the

majority of SI, AP, pitch, roll, and yaw shifts. Furthermore, due to the complexity of the head and neck area, internal organs risk an average of 2-3mm displacement error after initial setup (Wei et al., 2020). A study by Covington et al. (2019) reviewing patient setup using phantoms resulted in larger than expected RTDs, which increased as the location of the isocenter increased in distance from the reference surface. These studies suggested regular calibration and the combined use of IGRT techniques to increase patient setup accuracy.

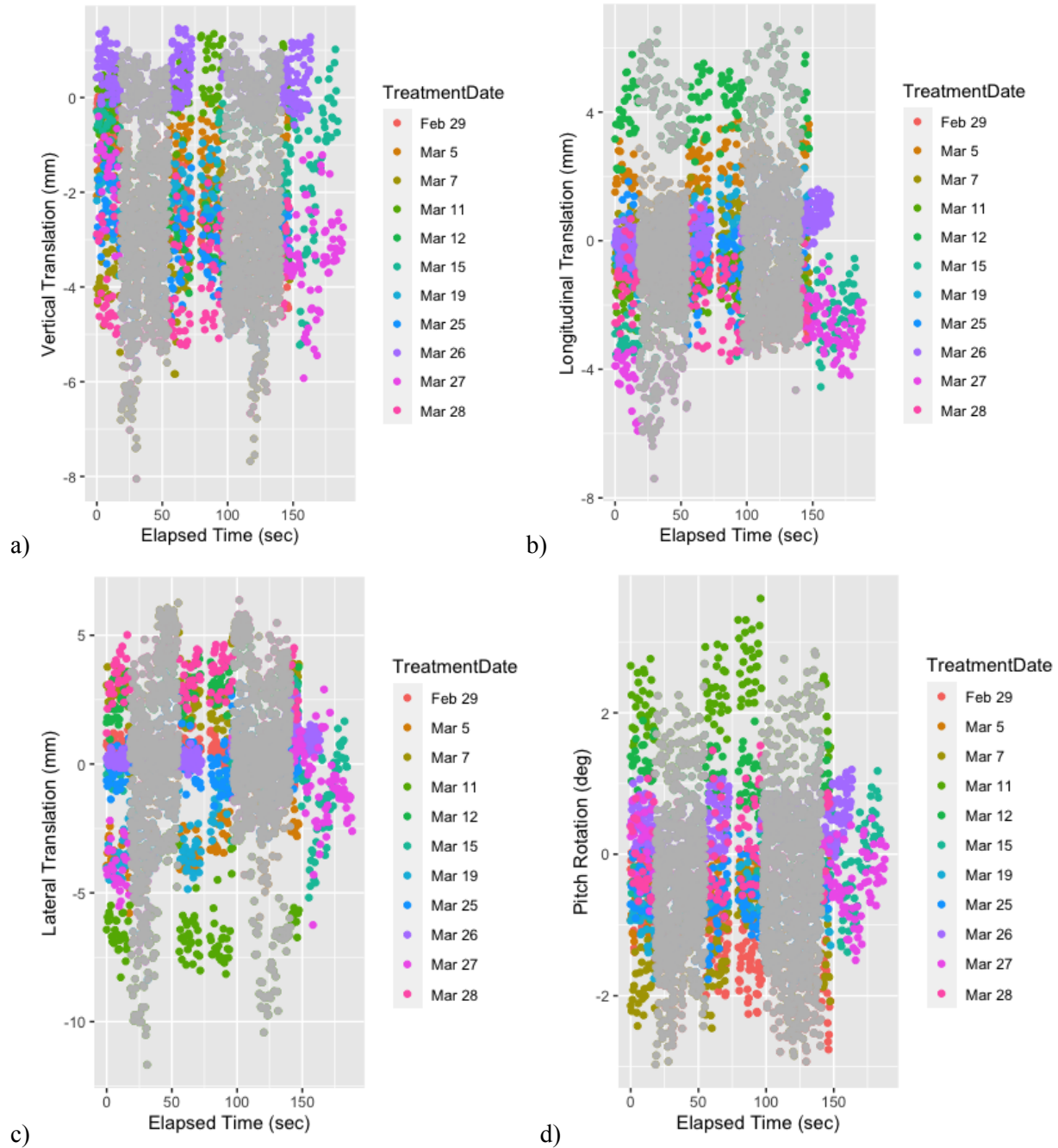
This study is limited by the small sample size, of only 4 patients due to initial difficulties in data collection and in using AlignRT, we are unable to use patients that have completed treatment already. Furthermore, the inclusion of a patient with a non-larynx malignancy and thus a slightly different isocenter location, coupled with treatment administered by a separate team of radiation therapists, introduces inter-patient uncertainties. These factors constrain both the internal and external validity of our findings.. Additionally, almost 50% of the data points sampled for intrafractional motion was excluded due to the AlignRT cameras tracking our ROI choice being shielded by the gantry at times, resulting in skewed information. Studies report that the ROI should be chosen to reflect the position of equipment surrounding the patient (Al-Hallaq et al., 2022), however in some cases this is not possible. Some studies have also chosen to solely exclude this skewed data (Covington et al., 2020; Qubala et al., 2022). The surface geometry of our ROI also posed challenges in validating interfractional data. This was primarily due to the planar and symmetrical nature of the clavicle surface we tracked. While target-volume based image registration (TV-IM) was performed by registered treatment planners at our institution, clavicle-based image registration (C-IM) was performed by an undergraduate student. While C-IM was primarily registered through automatic image registration functions in Eclipse, the lack of significant expertise could have also introduced slight uncertainties. Additionally, our ability to assess shoulder and body positioning below the neck was constrained by the limited range of the cone beam CT machine in the inferior direction. Consequently, we were unable to evaluate a broader area for both surface monitoring and image registration data, constraining our range of interfractional and intrafractional motion analysis.

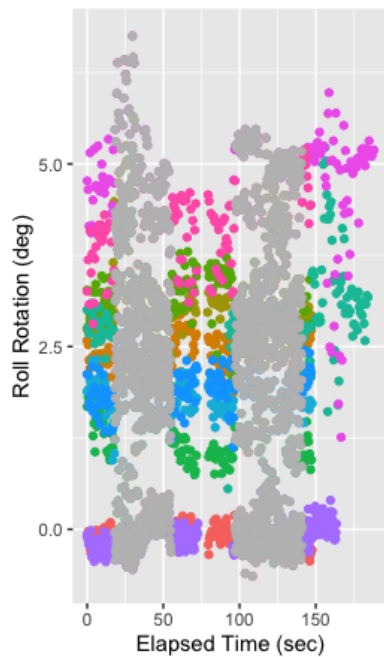
Conclusion

Currently, the data obtained from our study suggests that there is some shoulder movement based on the clavicle positioning relative to target volume area immobilization. Both intrafractional and interfractional movement seems to remain within the PTV margins of our institution, with a few outliers. Immobilization of the shoulders has demonstrated adequate accuracy using a shoulder cantilever depression system for larynx cancer patients undergoing VMAT. Alternatively, our observations of the accuracy of AlignRT in tracking the supraclavicular region seems to be much lower than what the literature suggests. Utilizing AlignRT for patient setup in our study was not feasible for reproducing patient shoulder positioning due to our ROI limitations, thus parameters influencing surface-guided systems should be evaluated carefully in future studies. Further investigations using a larger sample size may be useful in the external validity of our results.

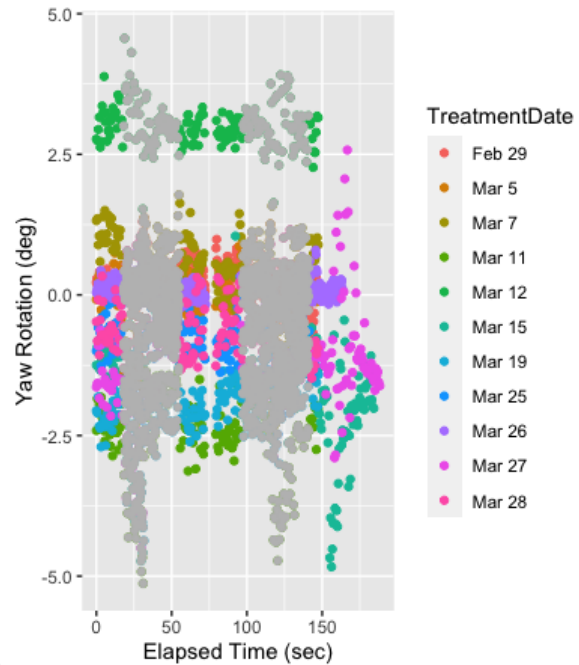
Appendix

Appendix 1. Example of translations and rotations as recorded by AlignRT across the length (2 arcs) of Patient D's treatment fraction. Data is recorded relative to the DICOM planning CT as the reference surface. Data points shaded in grey represent the sections where the gantry blocks the collection of representative and accurate surface monitoring. Coloured data points represent the data used for intrafractional motion results, with each colour representing a different treatment fraction. a) Vertical (AP) translations; b) Longitudinal (SI) translations; c) Lateral (LR) translations; d) Pitch rotations; e) Roll rotations; f) Yaw rotations.



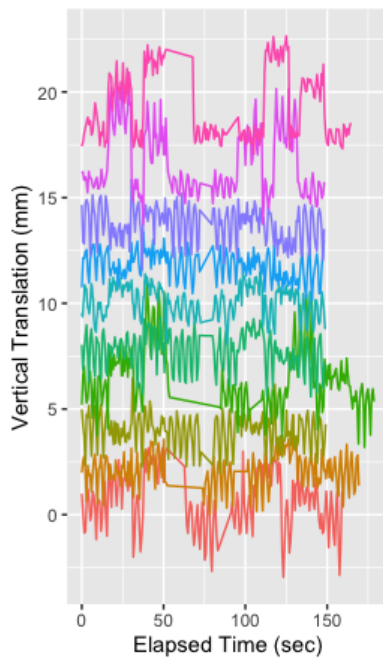


e)

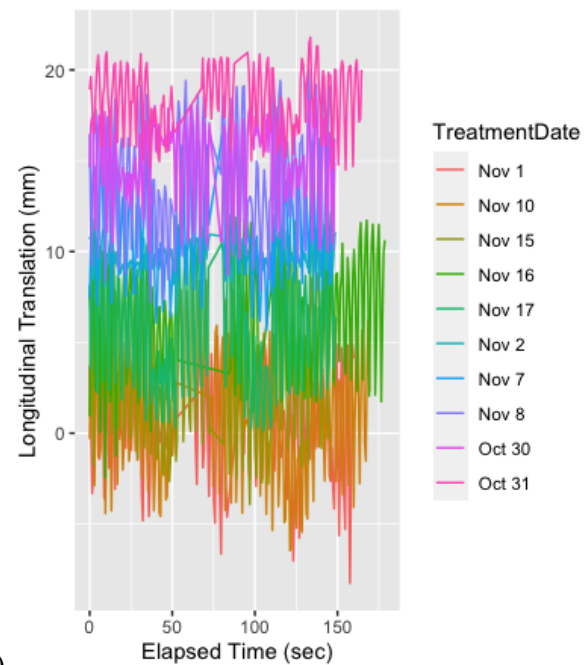


f)

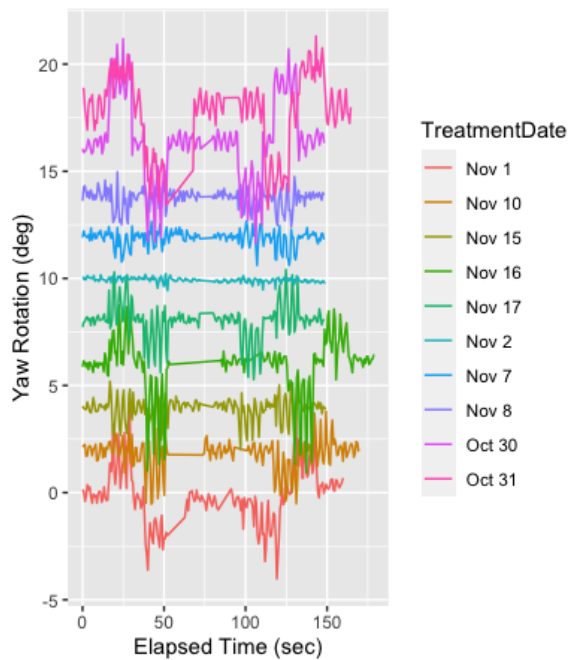
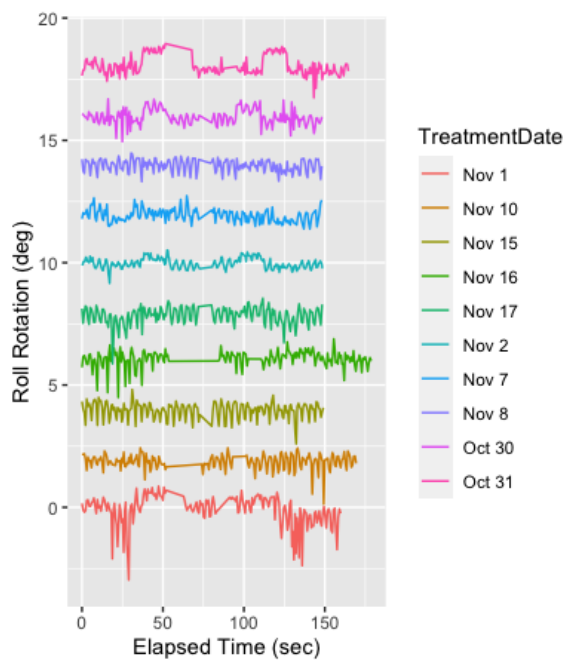
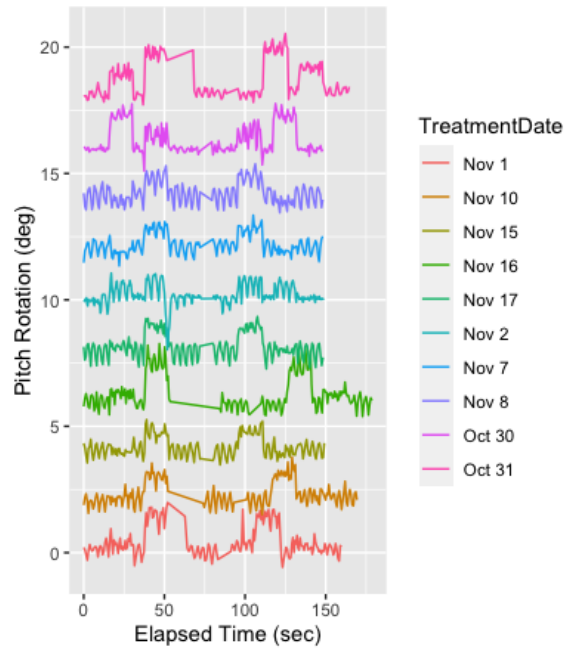
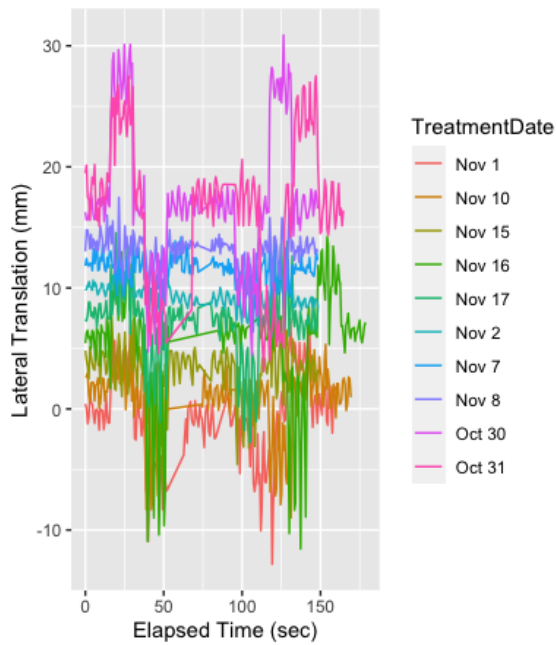
Appendix 2.a) Raw AlignRT data collected from Patient A log files. Data from each treatment date is represented by a coloured line, offset by 2mm between each date. Data is recorded relative to the average of the first 10 points where the treatment beam was active for each treatment fraction. Consistent, thin peaks and valleys represent breathing cycles of the patient. Flat regions across all patients exemplify gaps of time where the radiation beam was turned off due to patient difficulties or delays between arcs. i) Vertical (AP) translations; ii) Longitudinal (SI) translations; iii) Lateral (LR) translations; iv) Pitch rotations; v) Roll rotations; vi) Yaw rotations.



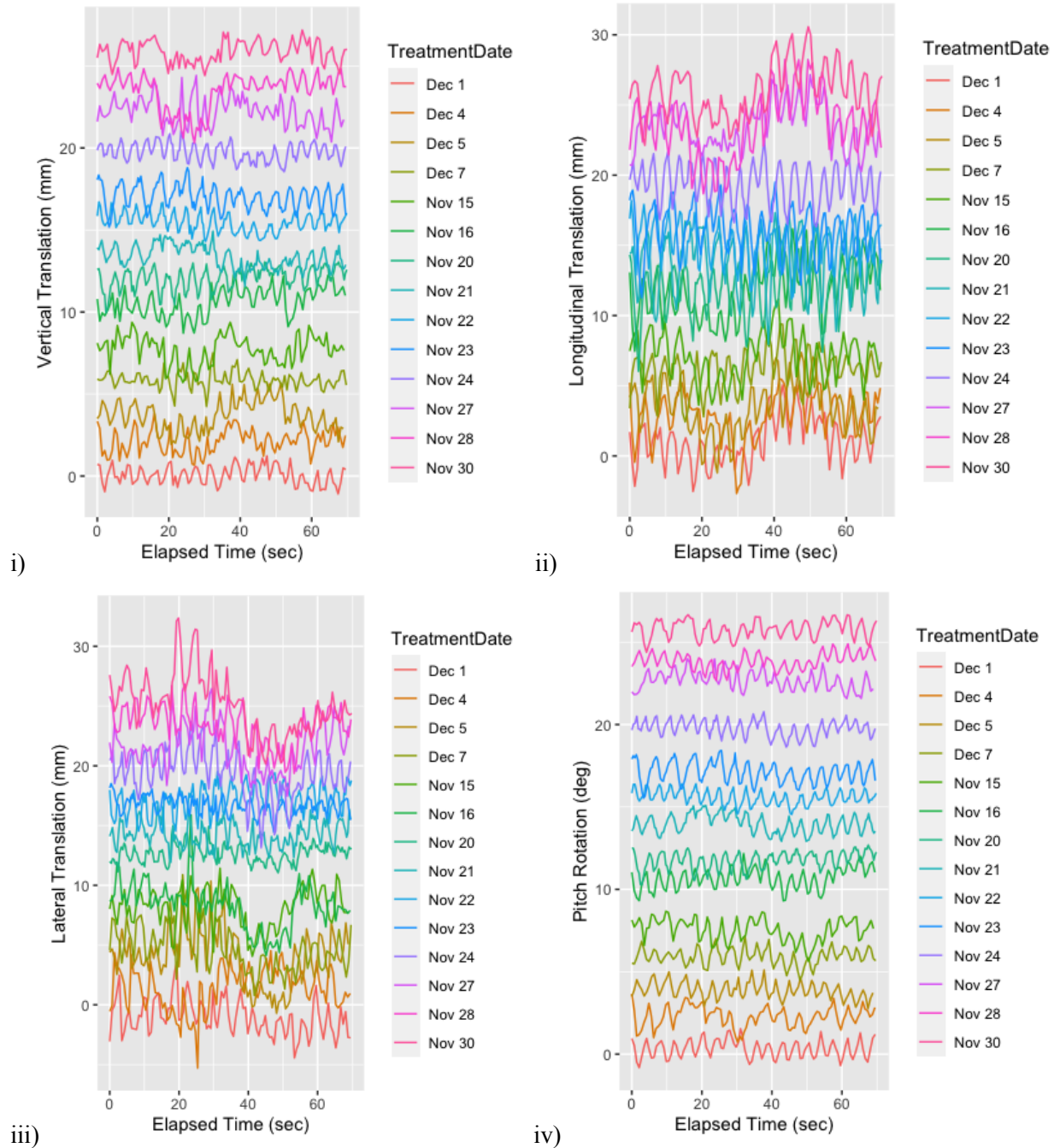
i)

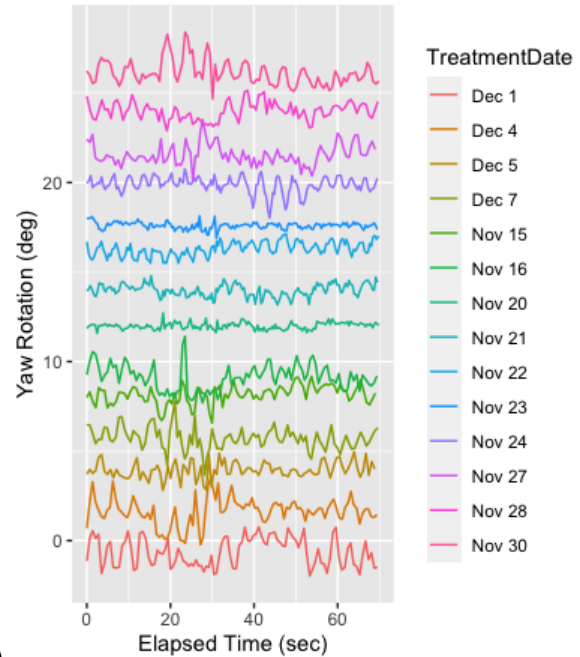
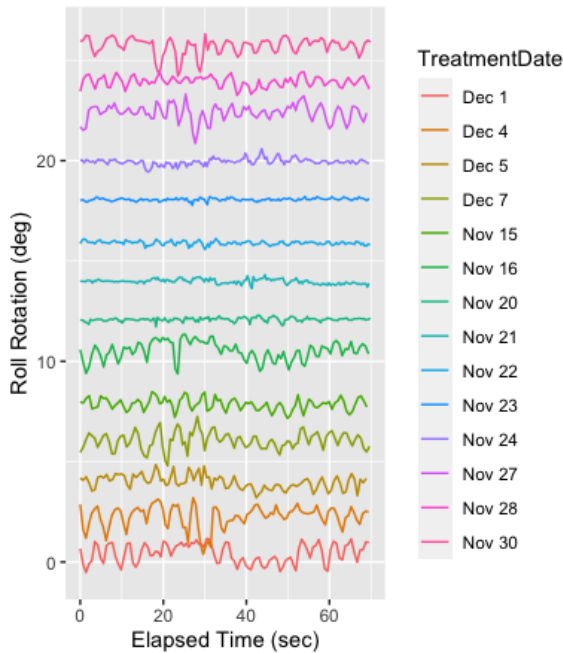


ii)

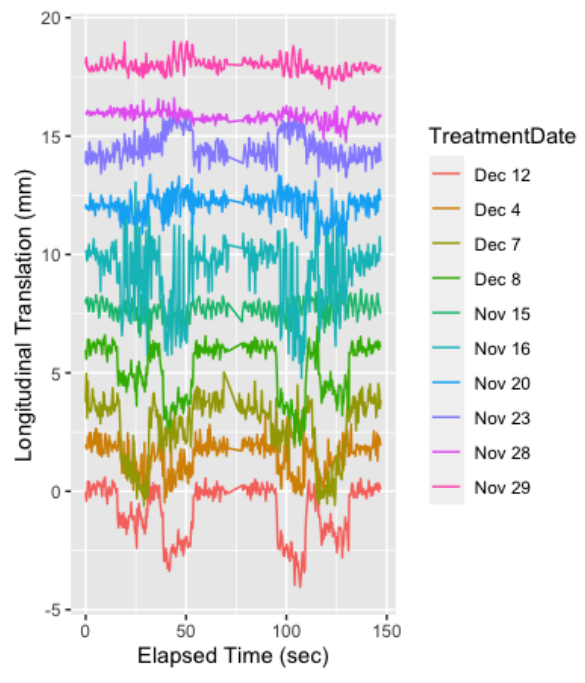
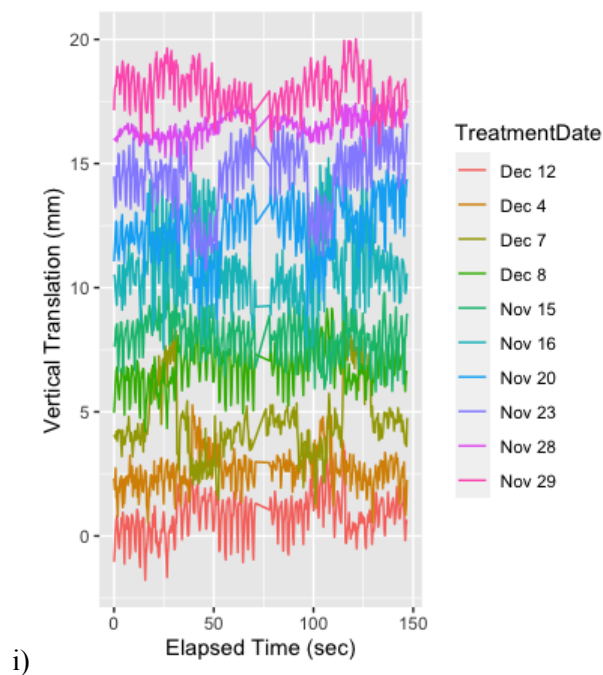


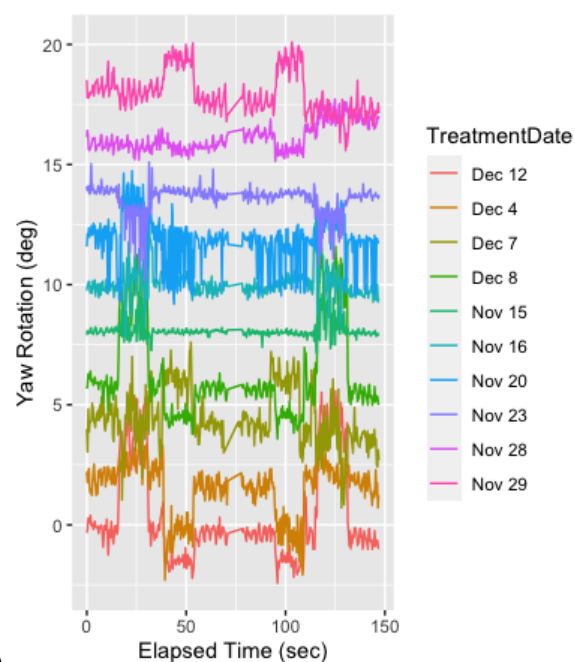
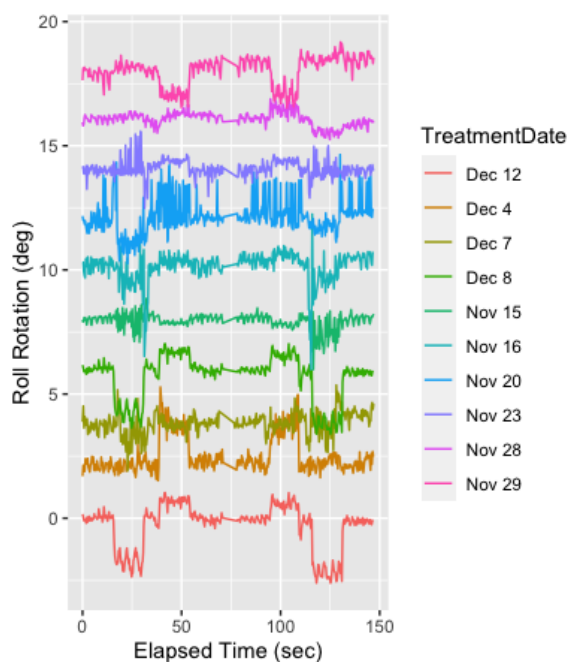
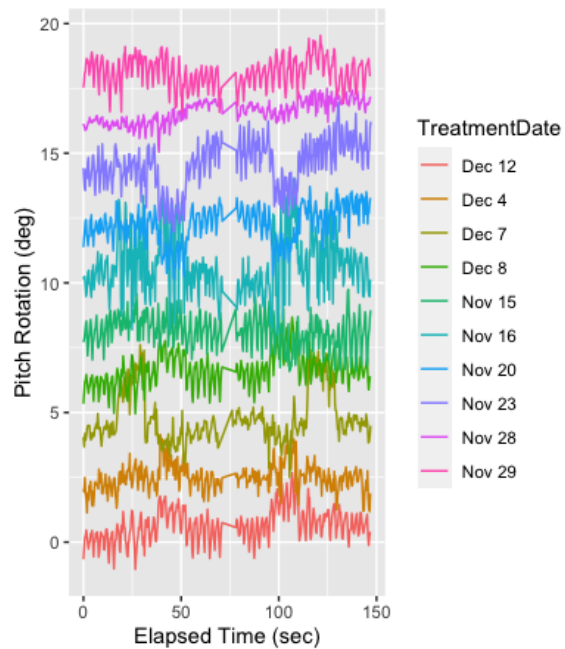
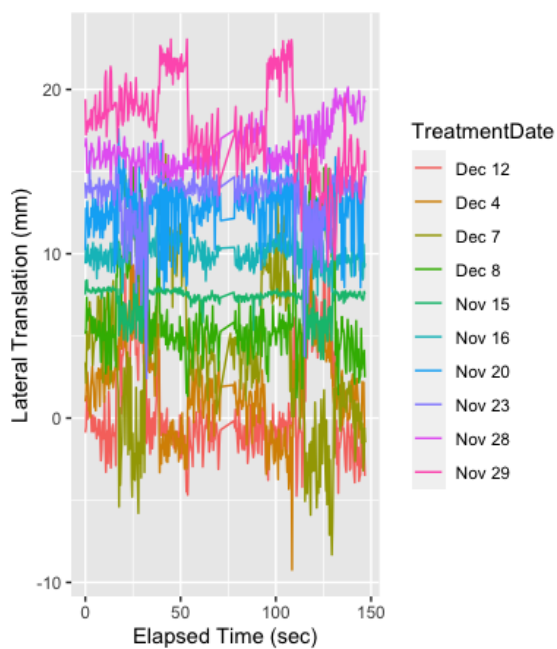
Appendix 2.b) Raw AlignRT data collected from Patient B log files. Data from each treatment date is represented by a coloured line, offset by 2mm between each date. Data is recorded relative to the average of the first 10 points where the treatment beam was active for each treatment fraction. Consistent, thin peaks and valleys represent breathing cycles of the patient. Flat regions across all patients exemplify gaps of time where the radiation beam was turned off due to patient difficulties or delays between arcs. i) Vertical (AP) translations; ii) Longitudinal (SI) translations; iii) Lateral (LR) translations; iv) Pitch rotations; v) Roll rotations; vi) Yaw rotations.



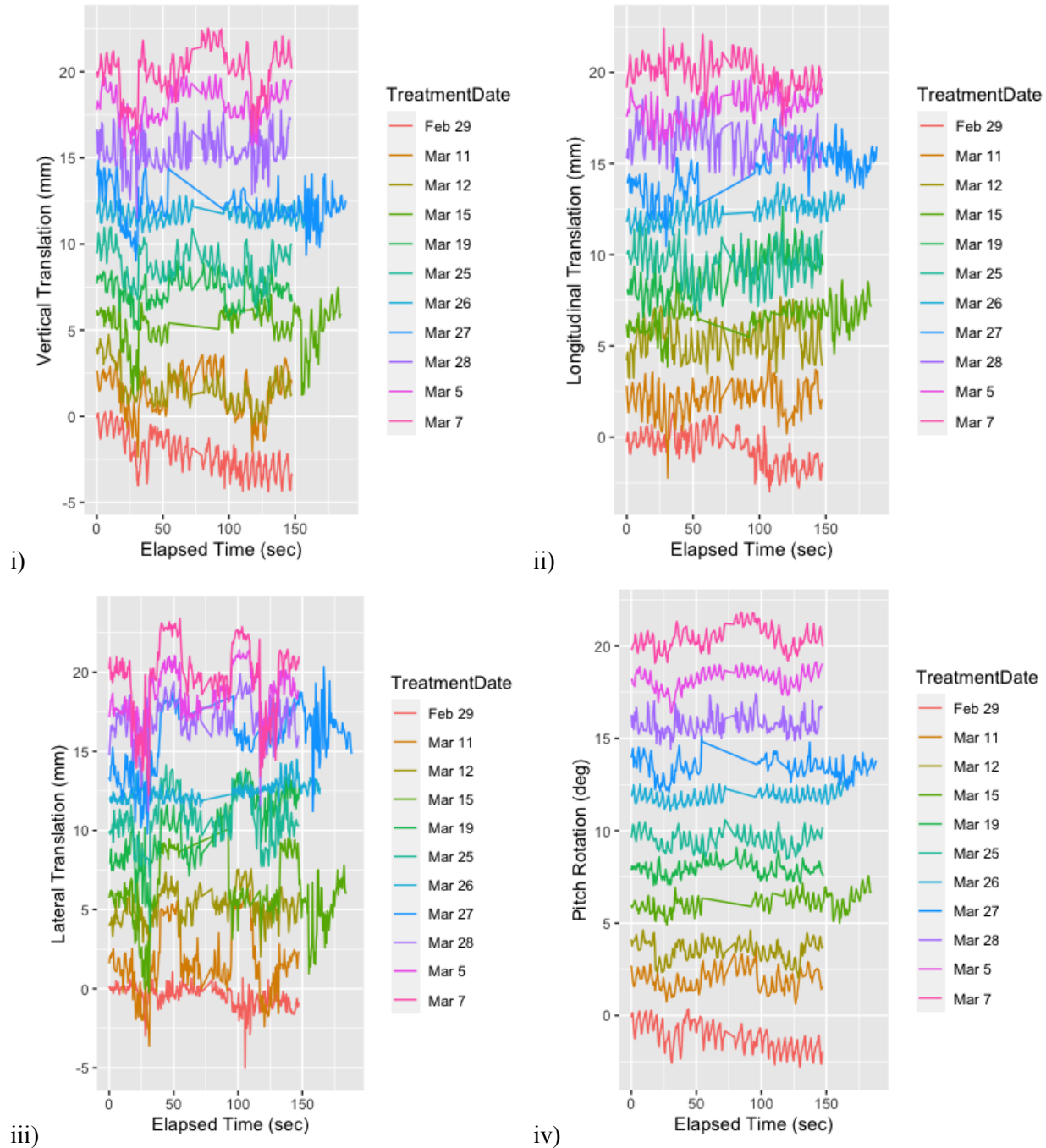


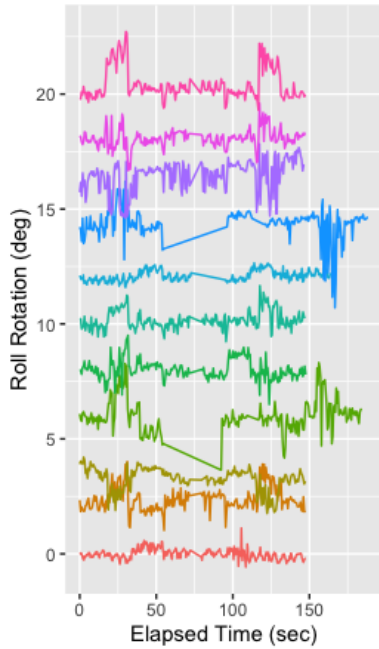
Appendix 2.c) Raw AlignRT data collected from Patient C log files. Data from each treatment date is represented by a coloured line, offset by 2mm between each date. Data is recorded relative to the average of the first 10 points where the treatment beam was active for each treatment fraction. Consistent, thin peaks and valleys represent breathing cycles of the patient. Flat regions across all patients exemplify gaps of time where the radiation beam was turned off due to patient difficulties or delays between arcs. i) Vertical (AP) translations; ii) Longitudinal (SI) translations; iii) Lateral (LR) translations; iv) Pitch rotations; v) Roll rotations; vi) Yaw rotations.





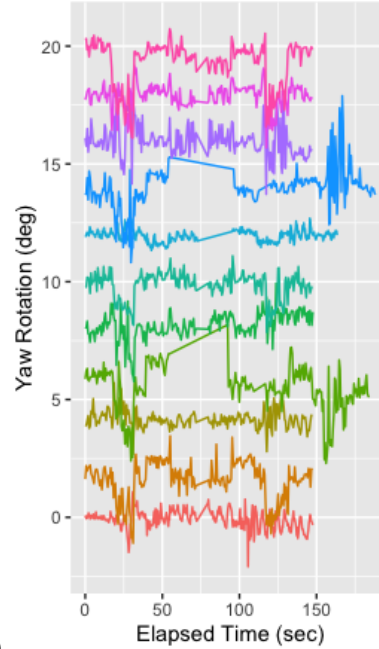
Appendix 2.d) Raw AlignRT data collected from Patient D log files. Data from each treatment date is represented by a coloured line, offset by 2mm between each date. Data is recorded relative to the average of the first 10 points where the treatment beam was active for each treatment fraction. Consistent, thin peaks and valleys represent breathing cycles of the patient. Flat regions across all patients exemplify gaps of time where the radiation beam was turned off due to patient difficulties or delays between arcs. i) Vertical (AP) translations; ii) Longitudinal (SI) translations; iii) Lateral (LR) translations; iv) Pitch rotations; v) Roll rotations; vi) Yaw rotations.





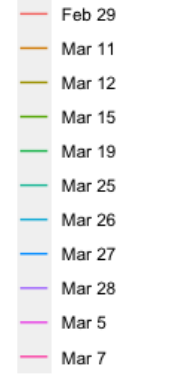
v)

TreatmentDate

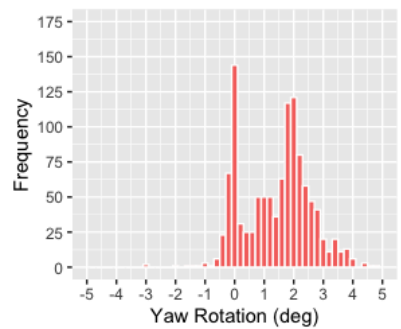
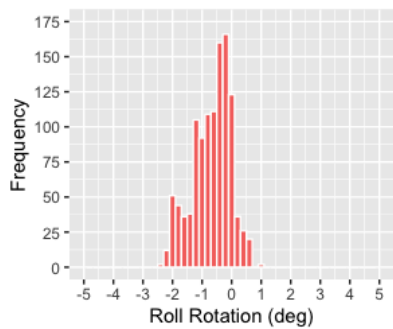
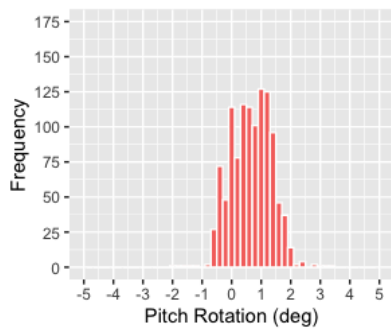
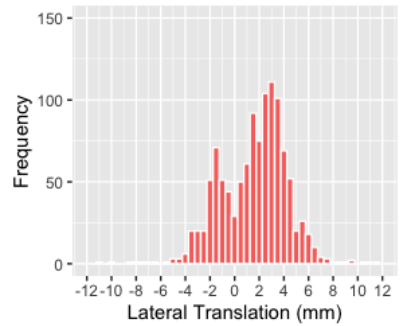
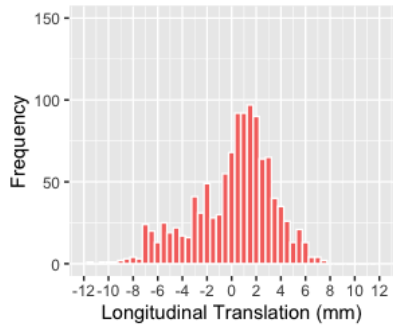
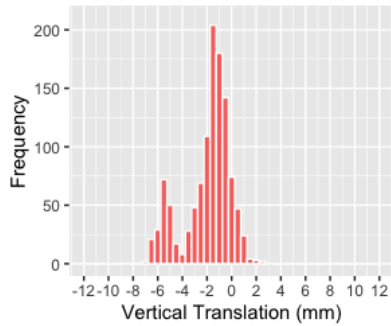


vi)

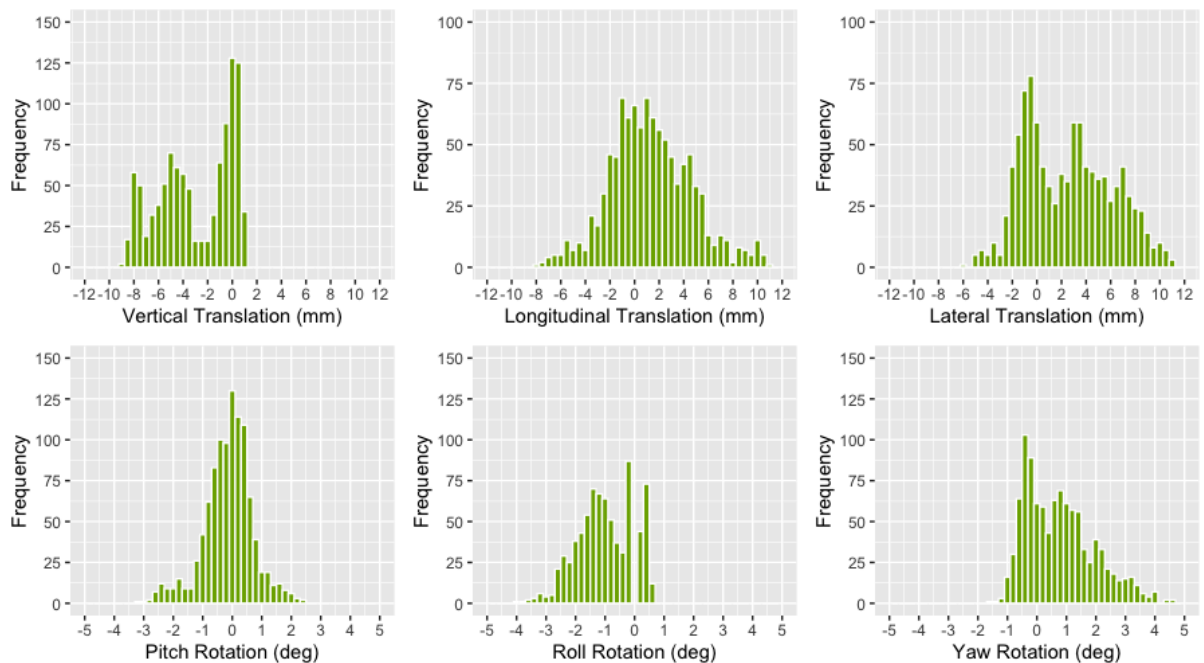
TreatmentDate



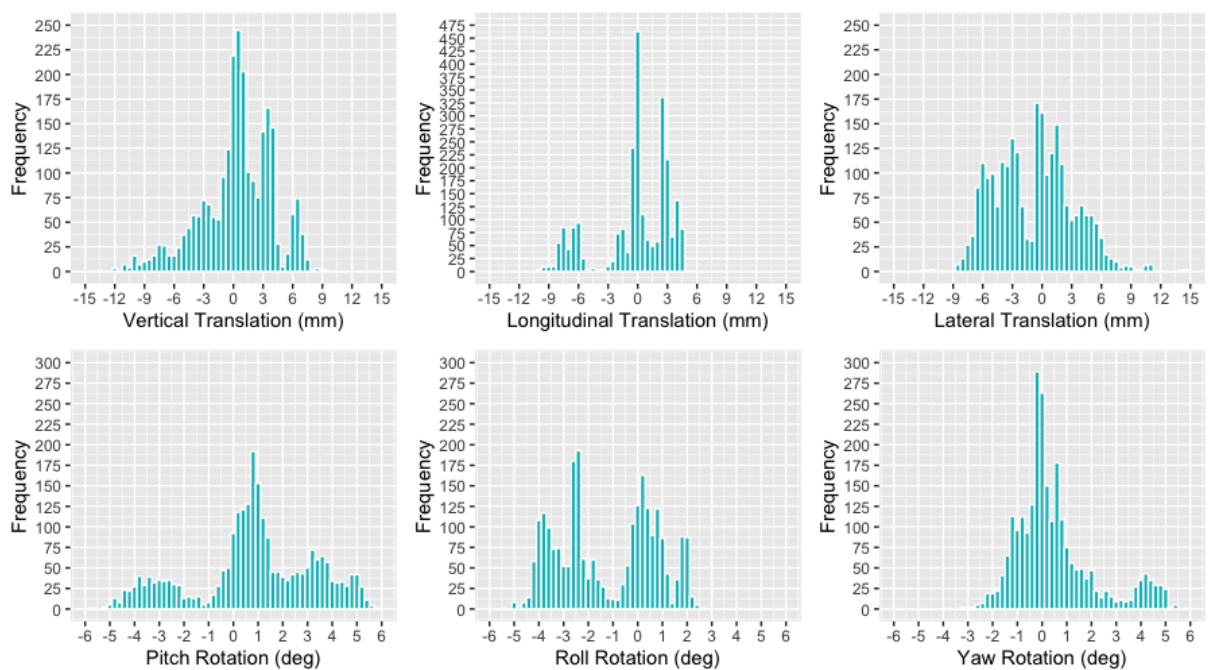
Appendix 3. Histograms of patient translational and rotational shifts as determined by AlignRT data. Each count of frequency corresponds to a single data point within the AlignRT dataset, recorded upon detecting movement in any direction. a) Patient A; b) Patient B; c) Patient C; d) Patient D.



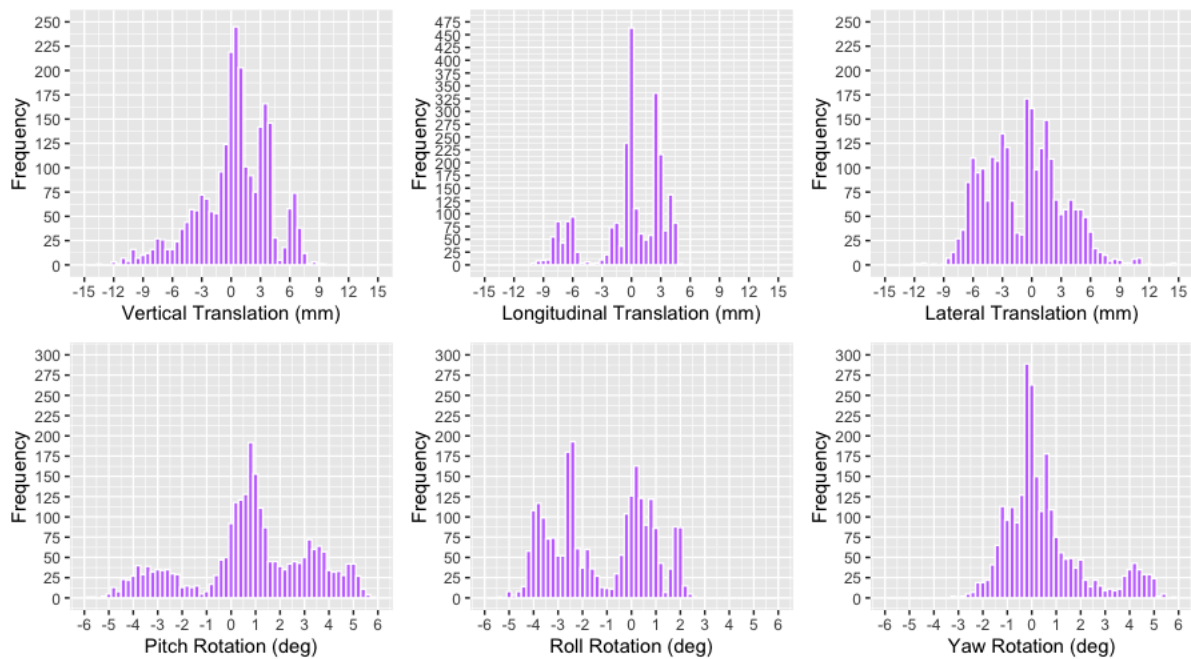
a)



b)

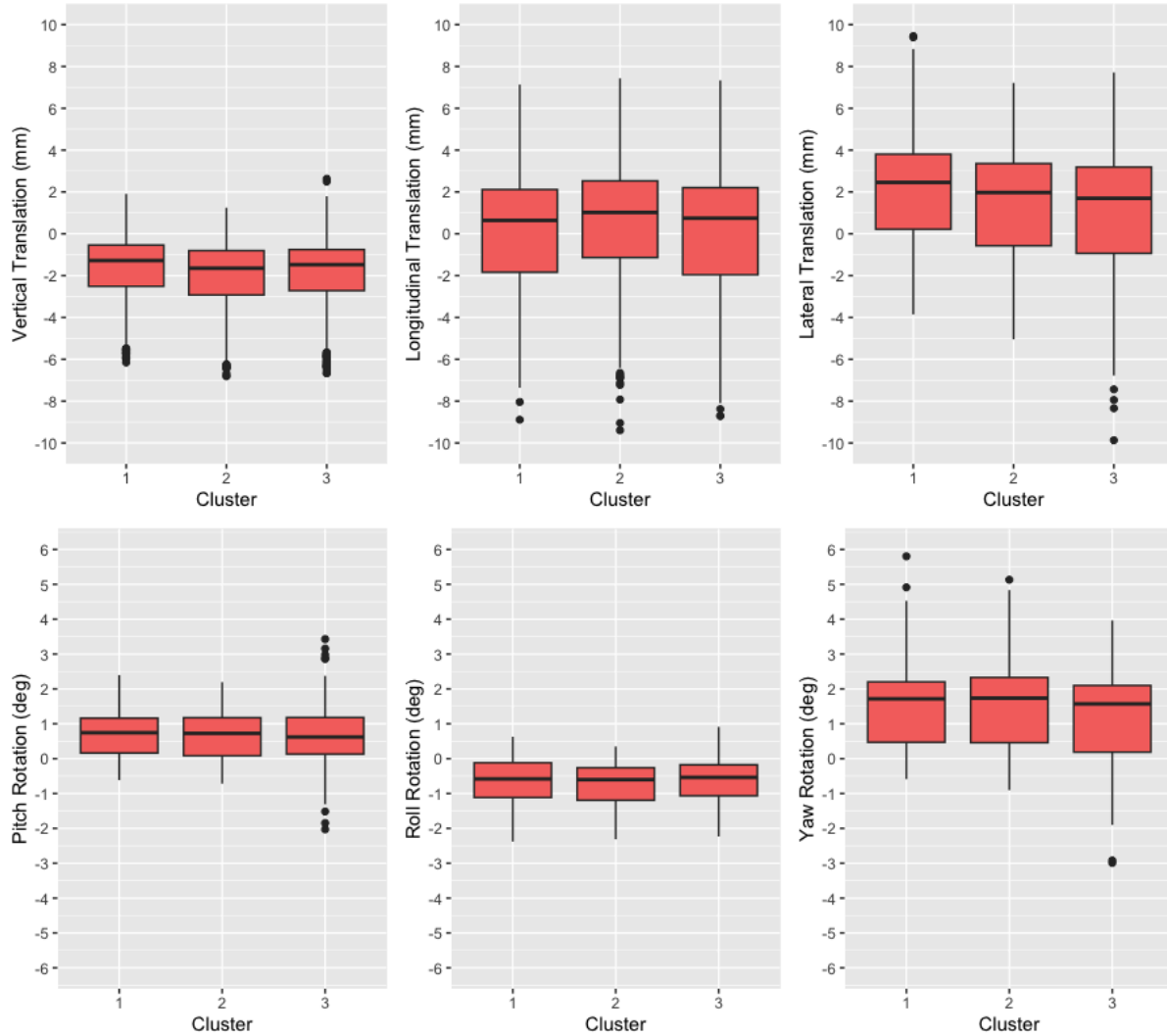


c)

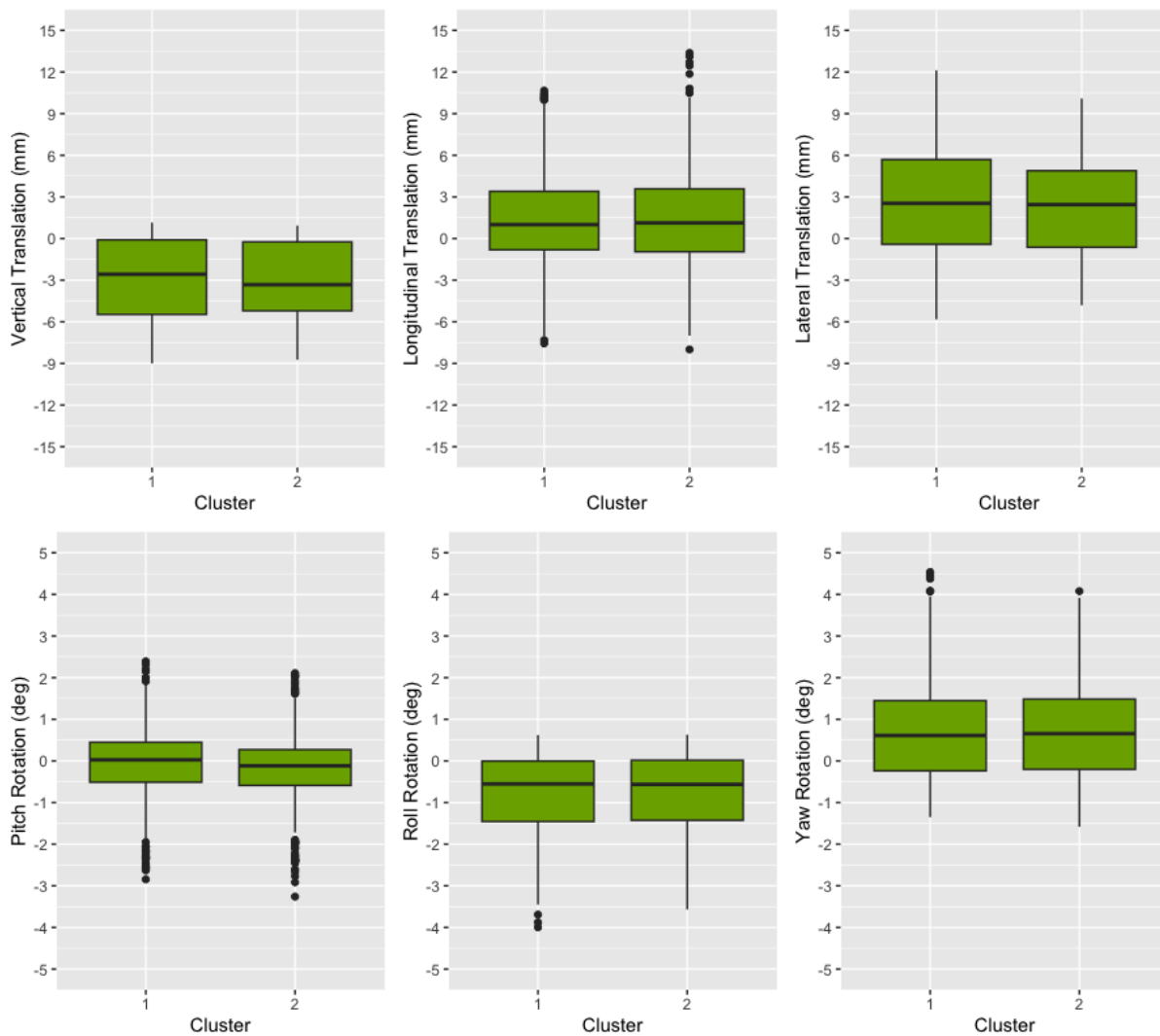


d)

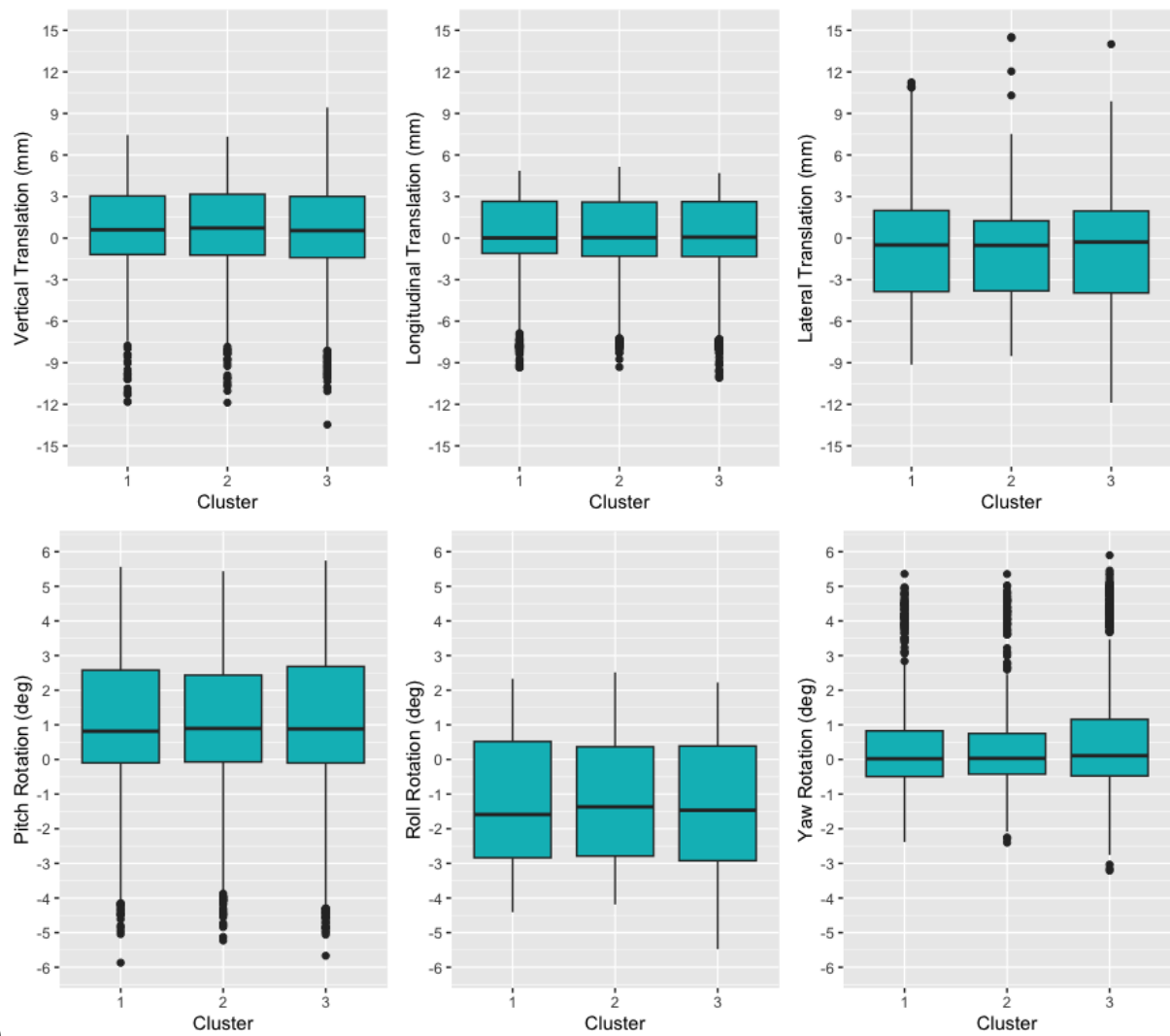
Appendix 4. Boxplots of patient translational and rotational shifts as determined by AlignRT data, organized by clusters. Clusters represent different gantry angle positions during both arcs of a patient's treatment. Cluster 1: CW 180° to 270°, Cluster 2: CW 90° to 180° and CCW 180° to 90°, Cluster 3: CCW 270° to 180°; CW: clockwise, CCW: counter-clockwise. Each data point corresponds to a single data point within the AlignRT dataset, recorded upon detecting movement in any direction. a) Patient A; b) Patient B; c) Patient C; d) Patient D.



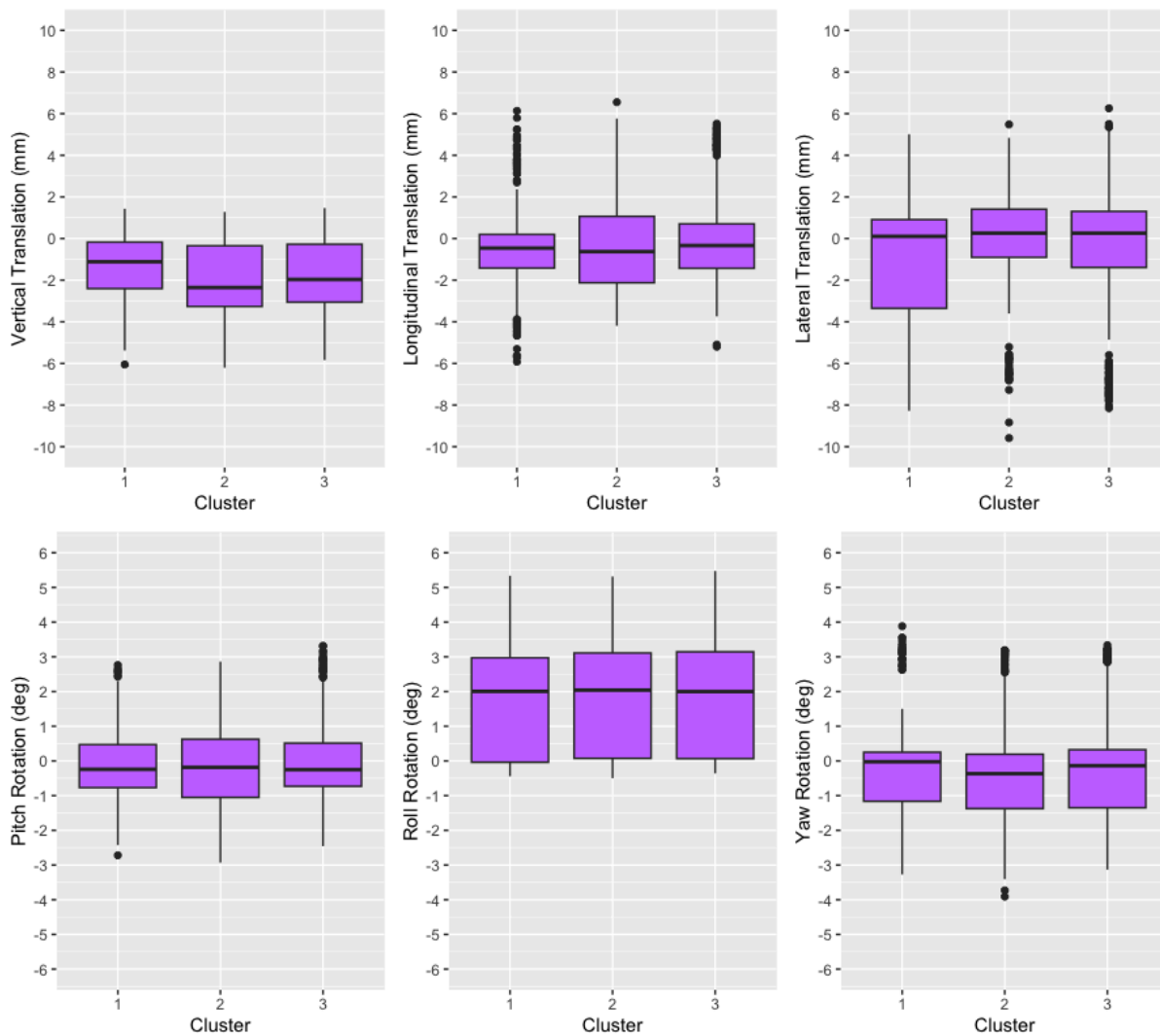
a)



b)



c)



d)

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